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LLM-Powered Dashboard using Multi-Agent Framework

1211101583

LUQMAN HAKIM BIN NOORAZMI

BACHELOR OF COMPUTER SCIENCE (HONS.)

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LUQMAN HAKIM BIN NOORAZMI

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Name of candidate: LUQMAN HAKIM BIN NOORAZMI

Faculty of Computing & Informatics

Multimedia University

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Abstract

Effective data analysis and interpretation are made possible by data visualization, which is essential for breaking down complicated datasets. There are many data visualization tool available for deriving analysis from raw data into insightful visual. However, a lot of the data visualization tools available today have a high learning curve, which limits their use for non-technical user. This limitation hinders the usability for non-technical user as more complex tasks require more time and resources to achieve the desired visual. There has been attempts by these existing data visualization tools to improve their platform's usability. This include implementing interactive and easy-to-use features, utilizing natural language to process data, and integrating AI assistant. This paper proposes a Natural Language Processing (NLP) and Large Language Model (LLM)-oriented solution to the usability limitation from existing data visualization tools by integrating LLM with the steps taken to produce data visualization and data analysis. This solution is aimed to reduce the number of steps and actions taken by the user to generate data visualization and data analysis. By incorporating text-to-SQL and LLM agents to automate data interpretation and visualization, this paper seeks to implement these solutions into a data visualization tool and assess the effectiveness of the LLM in producing accurate data visualization with minimal number of steps. The project will assess features from existing data visualization tools to gather the required features that makes up a data visualization tool and integrates it with LLM. The proposed data visualization tool will be able to receive prompt from user where a Structured Language Query (SQL) query will be generated by the text-to-SQL based on user prompt and data structure. With the query generated, it will be executed to retrieve related data from data source. LLM will convert the data into a structured format before making the best visualization choice. The LLM will also be capable of generating in-depth analysis of the queried data.

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List of Abbreviations/Symbols

Word/Phrase	Abbreviation
Artificial Intelligence	AI
AI Based Operating System	AIOS
Bidirectional Encoder Representations from Transformers	BERT
Etcetera	etc.
Final Year Report	FYP
Generative Pre-Training Transformer	GPT
Identification	ID
JavaScript Object Notation	JSON
Large Language Model	LLM
Model-view-controller	MVC
No Date	n. d.
Natural Language Generation	NLG
Neutral Language Model	NLM
Natural Language Processing	NLP
Natural Language Understanding	NLU
Pre-trained Language Model	PLM
Requests Per Minute	RPM
Rate per Minute	RPM
Software Development Life Cycle	SDLC
Small Language Model	SLM
Structured Query Language	SQL
Task Planning and Tool Usage	TPTU
User Acceptance Test	UAT
Uniform Resource Locator	URL

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Chapter 1: Introduction

1.1. Overview

Data visualization tools are essential in data analysis as they transform large and complex datasets into clear and interpretable visual representations. Over time these tools have evolved from basic charts and maps to advanced interactive displays that makes it easier to understand and analyze information. Today, they play a crucial role in data-driven decision-making with a variety of options available that offer unique strengths and limitations (Lavanya et al., 2023).

With the rapid progress of artificial intelligence (AI) Large Language Models (LLM) have become a key part of modern technology. Models such as ChatGPT and GPT-4 from OpenAI are known for their natural language understanding and reasoning abilities. They can handle a wide range of tasks and significantly improve human-computer interaction (Shen, 2024).

1.2. Problem Statement

Some data visualization tools have difficult learning curve, which means the data visualization tool gets harder to use when the task gets more complex. Due to the steep learning curve, the tool’s usability becomes a concern because it is restricted to users with certain level of technical skills. This hinders users with limited technical skill from being able to fully utilize the tool despite being rich of useful features.

1.3. Project Objectives

This project aims to address the limitations of current data visualization tools by integrating LLM. It leverages subdomains such as text-to-SQL that utilizes natural language process (NLP) and LLM agents. Text-to-SQL will be implemented to generate SQL query based on user prompt, and the use of LLM is to automate the visualization process. By analyzing existing visualization systems this project aims to design an enhanced data visualization tool that incorporates AI-driven automation to improve the usability of data visualization tool. By the end of the project, a prototype of the proposed data visualization tool will be developed and the result from the LLM will be assessed.

1.4. Project Scope

The size of the project is considered as small as it only aims to reuse concepts from existing systems and improving it using the proposed solution of LLM. The technology required mainly revolves around existing open source LLM and a framework that supports the language model. In addition, the use of text-to-SQL and LLM agents will be used as part of leveraging LLM and NLP into the data visualization tool.

1.5. Project Limitation

Based on the scope defined for the project, the limitations include time constraint. In a 14-week time frame mixed with other commitment, the depth of the analysis and implementation might be restricted. Some features that require more than implementing existing resources may not be feasible during this time frame.

Furthermore, open-source LLM may have its limitations compared to proprietary model, hence the result and response are reliance on its pre-trained knowledge.

1.6. Methodology

The traditional waterfall model of SDLC includes several phases of System Planning, Systems Analysis, System Design, System Implementation, and System Maintenance (Nurcahya et al., 2022). For this project, the phases include planning, analysis, design, and implementation. These phases will be spread out for two trimesters with FYP1 being the first part of the project and FYP2 being the second part of the project.

In this project, analysis on existing data visualization tools will be done to assess features that makes up a data visualization tool. In conjunction to the analysis, a questionnaire will be designed to understand which of the features are more likely to be implemented and which will be discarded. From the analysis, the design phase is to generate the logical design of the system with the help of Unified Modeling Language (UML) diagrams. With the baseline of the data visualization tool identified, a prototype of the data visualization tool will be implemented where the solution is to be integrated within the prototype.

The solution of integrating LLM with data visualization tool is proposed and will be developed within a 14-week time frame. Throughout the 14 weeks, a meeting will be held every two weeks to update the project's progress and plan the goal until the next meeting. Table 1.1 displays the task planned during the FYP1 project execution and description of the outcome of from each phase.

Table 1.1: FYP1 Tasks Description

Week Number	Task	Description
1	Problem Formulation and Project Planning & Background Study	The phase to define the problem statement, project objectives, and scope. Background research is conducted to review relevant literature, existing systems, and technologies. A timeline is created to guide project execution.
2		
3		
4		
5	Requirement Analysis	This phase focuses on gathering, analyzing, and documenting functional and non-functional requirements for the project.
6		
7		
8		
9	Design	In this phase, the system architecture, use case diagram, and sequence diagram are designed based on the requirements.
10		
11		
12	Prototype Development	Develop the prototype of the system based on the designed architecture and action flow.
13		
14	Submission	Project submission.

1.6.1. Milestone

Throughout these two parts of the projects and as part of the System Planning phase, a set of deadlines for the project's milestone is provided. This is to ensure the project will be ongoing and stay within its track until the end. Table 1.2 and table 1.3 displays the FYP1 and FYP 2 tasks with its proposed due date.

Table 1.2: FYP1 Tasks Deadline

Week	Task	Due Date
3	Project Planning	22/11/2024
5	Background Study	06/12/2024
7	Requirements Determination	20/12/2024
9	Requirements Structuring	03/01/2025
11	Design Specification Finalization	17/01/2025
12	Report Draft Completion	24/01/2025
13	Prototype Completion	31/01/2025
14	Report Completion & Submission	09/02/2025

Table 1.3: FYP2 Tasks Deadline

Week	Task	Due Date
1	FYP1 Review	28/03/2025
3	Software Module 1 Completion	11/04/2025
5	Software Module 1 Testing	25/04/2025
7	Software Module 2 Completion	09/05/2025
9	Software Module 2 Testing	23/05/2025
11	Software Development Completion	06/06//2025
12	Testing & Analysis of Result	13/06/2025
13	Software Refinement	20/06/2025
14	Report Completion & Submission	27/06/2025

Each milestone will be verified through regular meetings throughout the project's duration. The project planning phase is part of the SDLC's planning phase. The analysis phase includes activities of background study, requirements determination, and requirements structuring. Then, the design phase is executed to finalize the design specification. By this point of the project, the report draft is to be completed before finishing with the prototype. The report is then submitted before moving on to FYP2.

The FYP2 continues with the implementation phase that include the modules completion and testing. After the completion of software development, the refinement is done on the software based on the result of final testing. The project ends with report submission by the end of the trimester.

1.6.2. Gantt Chart

To visualize the progress for FYP1, the Gantt chart that depicts the project's progresses and phases throughout the trimester can be seen in figure 1.1.

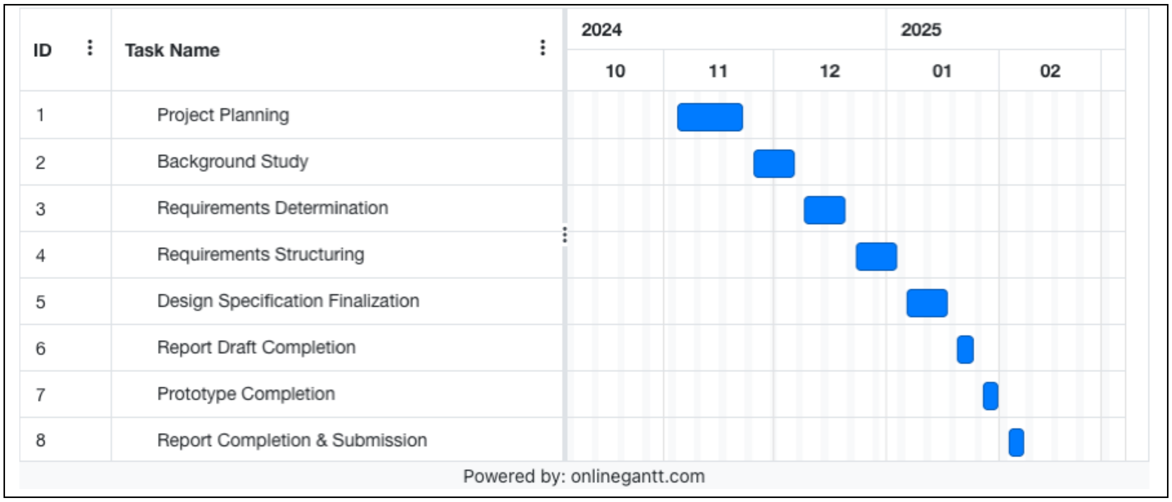


Figure 1.1: FYP1 Gantt chart

1.7. Target Audience

The target audiences are mainly university students, specifically computer science students, learning and research purposes. Data visualization tools user could benefit from this project's approach of visualizing data as the proposed tool is similar to other existing data visualization tools. LLM practitioner could benefit from the integration of LLM by assessing the implementation and outcome from the project.

1.8. Summary

In summary, data visualization often come with steep learning curves that limits accessibility for non-technical users. To address this challenge, this project aims to develop an end-to-end data visualization tool that integrates LLM to simplify complex user actions and enhance automated visualization generation.

The project methodology involves assessing existing visualization tools, conducting user research, designing a logical system architecture, and developing a prototype.

Chapter 2: Background Study

2.1. Overview

This section explores key technologies and concepts related to data visualization and its tools, text-to-SQL, and LLM agent. Data visualization aids in simplifying complex information by representing data visually through existing data visualization tools, while LLM is used to process and generate response in a human-like text. Autonomous agents enhance LLM’s ability to operate independently in explicit environments. NLP bridges the gap between human language and computers, and text-to-SQL systems further improve accessibility by allowing users to query structured databases using natural language. These advancements contribute to making AI-driven tools more intuitive and efficient across various industries.

2.2. Data Visualization

Data visualization is a method of displaying data in its visual representation. Data visualization tools aids data analysts in interpreting data and draw conclusions from it before making the final decision. Many data visualization tools have been developed with the goal of better observation to the data, which essentially lead to quicker and better decisions to be made. To achieve this goal, the visualization aspect becomes the main focus, as innovative techniques that affects the representativeness of the data is important, and this include primitives such as colours, elements, and dimensions, and analyses (Skender & Manevska, 2022).

There are many forms of visualizations data can be disguised as depending on the type of information being unveiled. Some form of visualization includes bar charts that are effective for comparing different categories or illustrating data distribution. Horizontal bar charts are beneficial for lengthy category label. Line charts are used to visualize trends over time, frequently employed in financial, scientific, and time-series data analyses by connecting data points with lines. Pie charts rely its effectiveness on limited number of categories for representing parts of a whole or composition of a dataset. Scatter plots make use the x-axis and y-axis as variables and plotting them against the data points as dots on the graph, making it valuable for spotting outliers, identifying clusters, and observing relationships between variables (Choudhary et al., 2024).

Data visualization tools of today exist with its own strengths and weaknesses. Some of the tools include Microsoft Power BI, which is best for business intelligence. Tableau is known for creating interactive charts while Qlik Sense excels in artificial intelligence-driven analytics. Choosing the right tool depends on specific needs as the field of data visualization continues to evolve. Data visualization tools have a long history, progressing from simple maps and charts to advanced interactive displays. Today they are essential for understanding complex information and making data-driven decisions. While there are many options for those entering the field, these tools also come with limitations (Lavanya et al., 2023). Table 2.1 displays the limitations of some existing data visualization tools and its limitations.

Table 2.1: Data visualization tools limitations

Data Visualization Tool	Limitation(s)	Cited Sources
Power BI	Difficult learning curve	Skender & Manevska (2022), Lavanya et al. (2023)
Tableau	No option to refresh report	Skender & Manevska (2022), Lavanya et al. (2023)
	No auto-scheduling when manually updating data	
Oracle Analytics Cloud	Limited graphics	Skender & Manevska (2022)
Google Data Studio	Difficult learning curve	Lavanya et al. (2023)
	Limited functionalities	
	Data processing limitation	
D3.js	Difficult learning curve	Lavanya et al. (2023)
	Complex set up	
	Tough for beginners	
Matplotlib	Difficult learning curve	Lavanya et al., (2023)
Plotly	No automation process	Lavanya et al. (2023)
DOMO	Not intuitive interface	Skender & Manevska (2022)

Based on table 2.1, various data visualization tools are proved to have different limitations. Power BI has a steep learning curve. Tableau does not have an option to refresh reports and does not support auto-scheduling for manual updates. Oracle Analytics Cloud has limited graphic options. Google Data Studio is difficult to learn and has limited functionalities. It also faces restrictions in data processing. D3.js requires a complex setup and is hard for beginners to use. Matplotlib is also difficult to learn. Plotly does not support automation and DOMO has an interface that is not intuitive.

Tableau promotes usability through some interactive and easy to use features by providing drag-and-drop feature, implemented a data viewer through Tableau Reader, allows data size manipulation, introduced interactive table, provide highlight and filter for data, and automatically display new features to name a few. These features are useful to user when generating data visualization.

Attempt to bridge the technical gap between non-technical user and data visualization tool can be further assessed through Oracle Analytics Cloud. Oracle Analytics Cloud is a data visualization tool for reporting and analytics in organizations of any size. It uses machine learning to detect important information through clear visualizations. One of its main benefits is processing data in natural language. Users do not need coding skills to explore and analyze data. The tool also supports multidimensional data analysis and scenario simulations. These features help provide a detailed view of business data even for non-technical user (Skender & Manevska, 2022).

M Cognos Analytics includes an AI Assistant that responds to inquiries in plain language. It suggests new data visualizations and joins to help users discover hidden relationships in data. The AI capabilities also assist enterprises in identifying patterns in customer behavior, market trends, and business performance. These insights help businesses make informed decisions with ease. The AI-powered features enhance self-service analytics, making data exploration more accessible to users (Lavanya et al., 2023).

From these examples of solutions to improve usability in data visualization tools, it can be summarized that tableau provides easy-to-use and interactive feature, Oracle Analytics Cloud leverages NLP for processing data, and M Cognos Analytics integrated AI Assistance to serve user in plain language by suggesting visualization, data relationship, and market trends for business decision. These attempts only enhance specific area of the data visualization tools, and the use of NLP are only targeted towards understanding the data.

2.3. Large Language Model

The development of language models has gone through four key stages. The first stage, Statistical Language Models (SLMs), emerged in the 1990s and used basic mathematical techniques to predict words based on past patterns. While helpful in early search engines and text-based applications, these models struggled to handle large amounts of data effectively. The second stage introduced Neural Language Models (NLMs) which used more advanced learning methods inspired by how the human brain processes language. These models could better understand relationships between words, making them more useful for applications like speech recognition and translation. The third stage saw the rise of Pre-trained Language Models (PLMs) which took a different approach by first learning from massive amounts of text before being fine-tuned for specific tasks. Models like BERT and GPT-2 made significant improvements in text understanding and became the foundation for many modern AI applications. The final stage, Large Language Models, built on this idea by significantly increasing the amount of data and computing power used for training. Models like GPT-3 and ChatGPT demonstrated new abilities, such as answering questions, writing human-like text, and reasoning through complex problems. This evolution has transformed language models from simple text predictors into powerful AI systems capable of assisting with a wide range of tasks, from customer service to creative writing and beyond. LLM is becoming a crucial part of modern system interaction due to the ongoing advancement in the Artificial Intelligence field. Models such as ChatGPT and GPT-4 from OpenAI have remarkable reputation for understanding natural language, which allows them to accomplish a variety of tasks and greatly improve human-computer interaction (Zhao et al., 2023).

While large language models (LLMs) are highly versatile, they face challenges when applied to specialized fields or tasks that require high accuracy and real-time decision-making. Their responses are based on patterns learned from vast amounts of data but they do not guarantee precision especially for tasks involving complex calculations or the latest information. This can lead to errors or misleading answers which often referred to as hallucinations. The main reason for these limitations is that LLMs are trained on past data. They cannot account for every real-time situation which makes it difficult for them to consistently provide fully reliable results in specialized areas (Shen, 2024).

2.4. Agent

To overcome the difficulty in specialized areas, LLM agents are utilized. Autonomous agents have been recognized as intelligent entities capable of accomplishing specific tasks by perceiving the environment, planning, and executing actions. Research done on agents have come up with various type of agents varying in purpose from general use to agent-based operating system. Table 2.2 displays the different agents that have been introduced and the process involved within the agents.

Table 2.2: Agent's Usage within different domains

Agent	Method	Advantage	Disadvantage	Cited Sources
Large Language Model-based AI Agents	Task Planning and Tool Usage (TPTU)	Sequentially Mimics Human Problem-Solving	Misunderstanding Output Formats	Ruan et al. (2023)
		Richer Contextual Understanding	Struggling to Grasp Task Requirements	
		Flexibility in Task Management	Endless Extensions	
		Improved Learning From History	Lack of Summary Skills	
AIOS (LLM-based AI Agent Operating System)	Application Layer for integrating native/non-native agents and Kernel Layer for LLMs integration	Resource Isolation & Management	Resource Management Challenges	Mei et al. (2024)
		Improved Agent Performance	Memory and Storage Constraints	
		Faster Execution Time	Security and Privacy Risks	
		Higher System Throughput	Tool Integration and Execution Conflicts	
		Scalability	Scalability Concerns	

The examples of agents from table depict agent's range of specificity. This proves the usefulness of agents in automating complicated tasks in specific fields to overcome the limitation of LLM.

2.4.1. Multi-agent Framework

By facilitating cooperation between agents with specialised functions, multi-agent systems enhance the capabilities of single LLM agents (Talebirad & Nadiri, 2023). They collaborate to finish tasks that are too big for one agent to do alone. Every agent contributes to a common objective and possesses special skills. Tasks requiring in-depth thought and creativity benefit greatly from this teamwork, which includes exercises like discussion and introspection.

Recent research has examined their potential in domains including reasoning in a debate environment (Liang et al., 2023), and teacher-student role play (Jinxin et al., 2023), demonstrating that multi-agent systems are capable of managing challenging real-world problems.

2.5. Natural Language Processing

Since not everyone is fluent in programming or machine-specific languages, NLP helps bridge this gap by allowing users to interact with computers using human language. NLP is a branch of AI and Linguistics designed to help machines interpret and process human-written text. Its primary purpose is to simplify human-computer interactions by enabling communication in natural language. NLP is generally divided into two key areas: Natural Language Understanding (NLU), which focuses on interpreting text, and Natural Language Generation (NLG), which involves producing human-like text. Linguistics, the study of language, includes aspects like phonology (sound), morphology (word formation), syntax (sentence structure), semantics (meaning), and pragmatics (contextual understanding). Theoretical linguist Noam Chomsky, known for his syntactic theories, played a major role in revolutionizing the field of language structure and understanding. Meanwhile, Natural Language Generation (NLG) involves creating meaningful sentences, phrases, or entire paragraphs from structured data (Khurana et al., 2023).

Many NLP research has been driven by computer scientists, but professionals from linguistics, psychology, and philosophy have also contributed to its advancements. NLP not only enhances computer capabilities but also deepens our understanding of human language (Khurana et al., 2023).

NLP has been applied into various areas like Machine Translation, Email Spam detection, Information Extraction, Summarization, Question Answering (Khurana et al., 2023). To implement NLP into these areas, several tasks are executed by NLP, as shown in table 2.3.

Table 2.3: NLP tasks description

NLP Task	Description
Automatic Summarization	Generating concise summaries from long texts.
Co-Reference Resolution	Identifying words that refer to the same entity in a sentence or document
Discourse Analysis	Studying how language is used in different social and conversational contexts
Machine Translation	Automatically translating text from one language to another
Morphological Segmentation	Breaking words into smaller meaning-based units
Named Entity Recognition (NER)	Identifying and classifying names, places, and organizations in text
Optical Character Recognition (OCR)	Converting handwritten or printed text into a machine-readable format
Part-of-Speech Tagging	Labelling words in a sentence based on their grammatical role

Nowadays, with technological advancements, LLM have enabled a wide range of NLP tasks to be performed within a single generative framework. Tasks such as mathematical reasoning, text summarization, machine translation, information extraction, and sentiment analysis can now be handled more efficiently, achieving impressive results. Furthermore, some LLMs can perform NLP tasks without requiring additional training data, and in some cases, they even outperform traditional models that have been fine-tuned using supervised learning (Qin et al., 2024).

2.6. Text-to-SQL

Text-to-SQL systems function as a bridge between human language and structured databases by converting natural language queries into executable SQL commands. This technology makes databases more accessible to non-experts, enabling them to retrieve relevant information without learning SQL syntax. As databases become more complex, manually searching through data becomes inefficient, making text-to-SQL a valuable tool for simplifying data exploration.

While text-to-SQL systems are useful, building reliable and accurate models is challenging. Linguistic Complexity and Ambiguity causes natural language to be vague or contain complex sentence structures which makes it difficult for the system to interpret correctly. To generate precise SQL queries, the system must understand database structures including tables, columns, and relationships. Since database schemas differ across industries, correctly mapping natural language to a database is challenging. Some queries require advanced SQL functions such as nested subqueries, multi-table joins, or window functions, which can be difficult for models to generate accurately, especially if they have not encountered such operations during training. And lastly, models that are trained on specific databases may struggle to perform well in entirely different fields due to differences in terminology, query styles, and database structures.

As organizations increasingly depend on relational databases to handle vast amounts of data, the need for efficient and user-friendly querying systems is more important than ever. According to the 2023 Stack Overflow Developer Survey, 51.52% of professional developers use SQL, making it one of the most commonly used programming languages. However, for non-technical users, SQL can be difficult to learn and use. Text-to-SQL helps eliminate this barrier by enabling natural language interactions with databases, allowing businesses to make faster and more data-driven decisions (Mohammadjafari et al., 2024).

2.7. Summary

To conclude the key findings from this section, data visualization is responsible for transforming raw information into visual formats such as charts and graphs, but the tools available suffers from high-level usability which limits the tools usage to only those with the technical skills. Existing data visualization tools have attempted to tackle this problem by utilizing NLP for non-technical user to easily interact with the tool but the solution is only applied on specific part of the tool rather than making a full-fledged NLP-oriented solution.

With LLM that have progressed from statistical models to large-scale AI systems like GPT-3 and ChatGPT it improves human-computer interaction but does not produce satisfying result in specific environment. AI agents, a solution to the LLM’s limit in specific settings, assists in automating specific tasks with systems like AIOS optimizing resource management and task execution.

NLP enables machines to process and understand human language, supporting applications like machine translation, sentiment analysis, and speech recognition. Text-to-SQL systems address the challenge of database querying by converting natural language into SQL commands that makes structured data more accessible to non-technical users.

Despite the data visualization tools advancements and evolution of LLM, these technologies continue to face challenges that hinders its full potential. The introduction of agents which is the result of LLM’s advancement and text-to-SQL which is the result of NLP’s advancement in recent time proves that NLP and LLM have evolved to overcome their own limitations.

In summary, these advancements from the domains mentioned in this section, agents and text-to-SQL, will used to assist in implementing the solution to the difficult learning curve problem that exists in existing data visualization tools.

Chapter 3: Requirement Analysis

3.1. Overview

The system is aimed to solve the problems identified earlier using the proposed solutions of text-to-SQL and multi-agent framework. Before getting started with the solution, the system that qualify as a data visualization tool must be developed first for these solutions to be integrated inside the system. This section will present the process of gathering requirements for the system through the mean of observing existing system and designing questionnaire, analyzing the results from the observation and questionnaire, and finalizing the functional requirements and quality requirements of the system.

3.2. Fact-Finding Techniques

3.2.1. Justification

To better understand the features that makes up a cutting-edge data visualization tool, two elicitation techniques were used: observation and questionnaire. While both aim to gather requirements, these two techniques offer different methods and steps to generate analysis that will be used to decide the final requirements of the system.

Through observation, analysis is derived from features that exist within the existing systems by understanding its usage amongst users (Gobov & Huchenko, 2021). Questionnaire on the other hand helps with understanding user's feeling towards the identified features (Kircher & Zipp, 2022). The features that were identified from the observation will be presented inside of the questionnaire to filter the features.

3.2.2. Observation

3.2.2.1. Preparation

Before starting the observation, some resources were prepared. First, the goal of the observation is established, which is to collect information regarding features from existing data visualization tools. The existing data visualization tools to be observed are amongst the four tools mentioned in the background study. The list of features available from these tools can be accessed through the respective tools' documentation.

Table 3.1 displays the description of the tools involved during the observation process.

Table 3.1: Existing data visualization tools feature description

Data Visualization Tool	**Description**	**Cited Sources**
Power BI	A data visualization tool that helps users transform data from different sources into interactive and reliable insights. It supports data from Excel files, cloud-based systems, and on-premises databases. Users can create business reports and dashboards to analyze data. The visualized information can be shared with others, making data exploration easier.	Biswal (2024) Microsoft Corporation, Inc. (2024)
Tableau	A comprehensive and adaptable data visualization tools that incorporates interactive visualization to harmonize, manage, and explore data integrated from various data sources.	Staff (2024) Tableau Software, LLC. (n.d.)
Looker Studio	A data visualization tool to create modifiable charts and tables with collaborative report to share insight to team member or broader audience. Reports can be personalized using pre-built template to present data from various data source.	Casarotto (2021) Google LLC. (2025)
Oracle Analytics Cloud	Helps business analysts and consumer with advanced, AI-driven self-service tools for data preparation, visualization, enterprise reporting, augmented analysis, and natural language processing.	Argano, (2022) Oracle Corporation, (n. d.)

3.2.2.2. Execution

With everything prepared, the observation process begins by gathering features from existing data visualization tools. Table 3.2 displays the list of features gathered across the different tools and its generalized description.

Table 3.2: Data visualization tools features

Feature	Description	Cited Sources
Data Connection	Connect to various data sources ranging from excel workbook, CSV, XML, JSON to relational database such as MySQL, Oracle, and db2.	Microsoft Corporation, Inc. (2024)
		Tableau Software, LLC., (n. d.)
		Google LLC. (2025)
		Oracle Corporation. (n. d.)
Report	Create report with visualization and data insight.	Microsoft Corporation, Inc. (2023)
		Google LLC. (2025)
		Oracle Corporation. (n. d.)
Chart/Visualization	Create suitable visualization for representing data	Microsoft Corporation, (Inc., 2023)
		Microsoft Corporation, Inc. (2025)
		Tableau Software, LLC. (n. d.)
		Google LLC. (2025)
		Oracle Corporation. (n. d.)

Dashboard	Create customized dashboard with selected visualization	Microsoft Corporation (Inc., 2025)
		Tableau Software, LLC. (n. d.)
		Oracle Corporation. (n. d.)
Collaboration	Collaborate with other user on a shared data	Microsoft Corporation (Inc., 2024)
		Google LLC. (2025)
Automated Visualization	Automatically generate data visualization based on the data	Tableau Software, LLC. (n. d.)
Advanced Analysis	Automatically generate analysis from data	Tableau Software, LLC. (n. d.)
Data Storytelling	Automatically generate story to present data	Tableau Software, LLC. (n. d.)
Data Modelling	Create data model to display relationships	Microsoft Corporation (Inc., 2024)
		Oracle Corporation. (n. d.)

3.2.2.3. Result

The result of the observation is presented in table 3.3 To grasp each feature's importance in a data visualization tool, comparison is done across various data visualization tools to see how frequent a feature exists amongst data visualization tools.

Table 3.3: Features' existence across data visualization tools

Features	Power BI	Tableau	Looker Studio (Google Data Studio)	Oracle Analytics Cloud
Data Connection	yes	yes	yes	yes
Report	yes		yes	yes
Chart/Visualization	yes	yes	yes	yes
Dashboard	yes	yes		yes
Collaboration	yes		yes	
Automated Visualization		yes		
Advanced Analysis		yes		
Data Storytelling		yes		
Data Modelling	yes			yes

Based on table 3.3, two features that stand out are Data Connection and Chart/Visualization as these exist in all four of the observed data visualization tools. Report and Dashboard exist in most of the tools, while features like Collaboration and Data Modelling only exist in half of the observed tools. The remainder, Automated Visualization, Advanced Analysis, and Data Storytelling only exist in only one tool.

3.2.2.4. Analysis

To summarize the findings from the observation, a feature's importance in a data visualization tool is measured based on how often it appears in various data visualization tools. It was found that Data Connection and Chart/Visualization are the most important, while Report and Dashboard are considered highly important in a data visualization tool. The remainder of the features are deemed useful but not as crucial and important.

3.2.3. Questionnaire

3.2.3.1. Preparation

With the features classified from the observation of its usage amongst users, the questionnaire is to help with analyzing its usefulness from user's perspective. To design a questionnaire, the project's aims, questions, and hypotheses need to be established (Kircher & Zipp, 2022).

The background study proposed the goal of leveraging LLM and text-to-SQL into data visualization tool. The questionnaire will help with answering question of what features makes up an up-to-date data visualization tool. This project is predicted to solve the problems of usability for non-technical user within existing data visualization tools.

With the baseline of the questionnaire in mind, the designing phase of the questionnaire begins. The questionnaire consists of both closed-ended question and open-ended question that starts with respondent demographic information. Afterwards, the questions are directed towards data visualization features and its usefulness. Lastly, questions on respondent opinions on data visualization tools will be asked towards the end of the questionnaire.

For closed-ended questions, multiple-choice questions are used on questions with multiple possible answer options. Since the answer choices may consists of too many options, an ‘other’ option is available when deemed necessary where respondent can specify any specific answer not listed as the options. The use of rankings is to order the answer in terms of frequency, proficiency, and usefulness of a subject.

Subjective answers are gathered through open-ended questions with the goal of understanding further the limitation and features of data visualization tools from respondent’s perspective.

The questionnaire consists of three pages that groups the questions based on the demographic, features, and personal opinions. The demographic section of the questionnaire is to gather the respondent’s background through questions relating to professional status, data visualization usage and proficiency, and experience with existing data visualization tools. For questions relating to the features, each feature’s usefulness from the observation process is ranked from a scale of 1 to 5 with 1 being least useful and 5 being very useful. The last section is designed to gather more insight into data visualization tools’ limitation and features the respondent wish to see in data visualization tools.

3.2.3.2. Execution

The questionnaire is distributed through shareable links using Microsoft Forms. A window of 10 days is reserved for responses from the questionnaire.

3.2.3.3. Result

A total of 26 response were collected. Figure 3.1 to figure 3.4 shows the response from the demographic section, figure 3.5 to figure 3.9 shows the response from the features section, and table 3.4 and table 3.5 shows the response from the open-ended section.

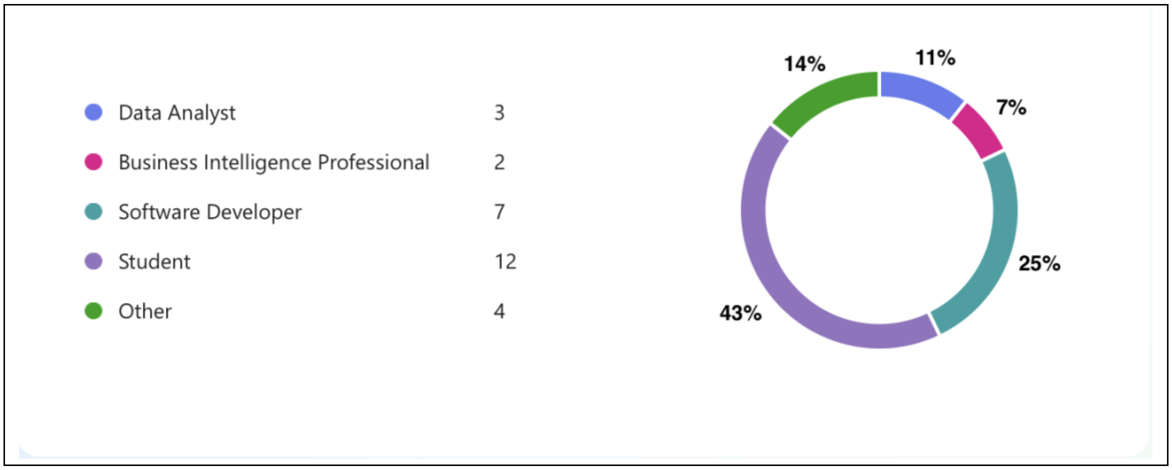


Figure 3.1: Respondents professional status

Most of the respondents are student and software developers. Few of the respondents are data analyst and business intelligence professional, while the ‘Other’ professional status is ‘Visual Editor’, ‘Customer Service’, and ‘Managers’.

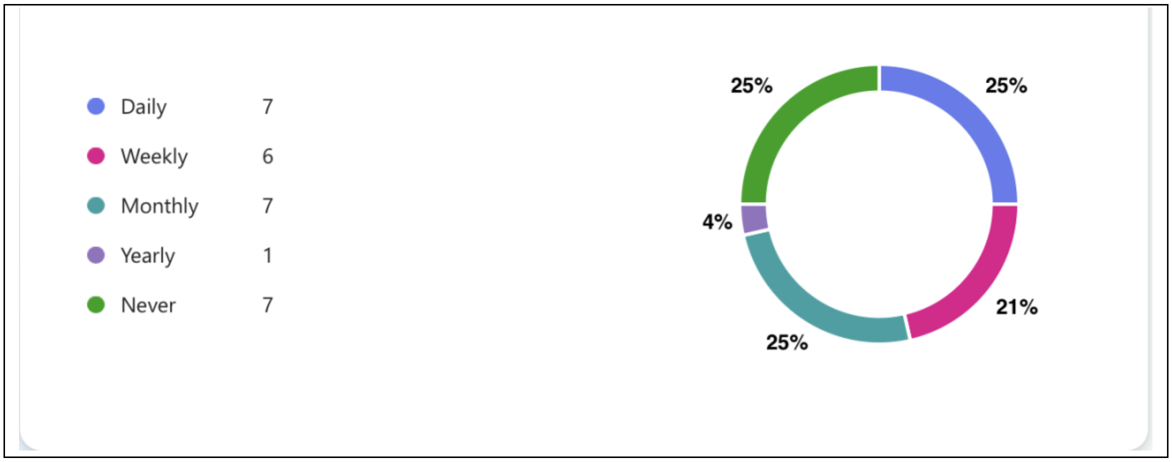


Figure 3.2: Respondents data visualization tools usage

The usage frequencies between daily, weekly, monthly and never are fairly balanced with only one respondent have the usage frequency of yearly.

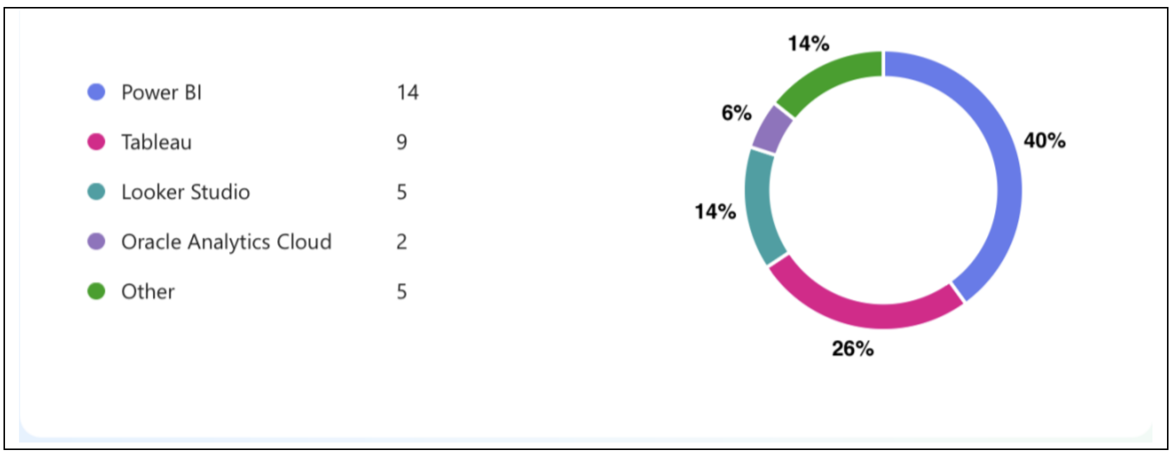


Figure 3.3: Data visualization tools used by respondents

Power BI stands out as the most popular data visualization tool amongst the respondent, followed up by tableau. Looker Studio and Oracle Analytics Cloud are in the minorities as both are rarely used amongst the respondent. The 'Other' data visualization tools are 'Jupyter Notebook', 'SPSS', 'Metabase', and 'Kibana'.

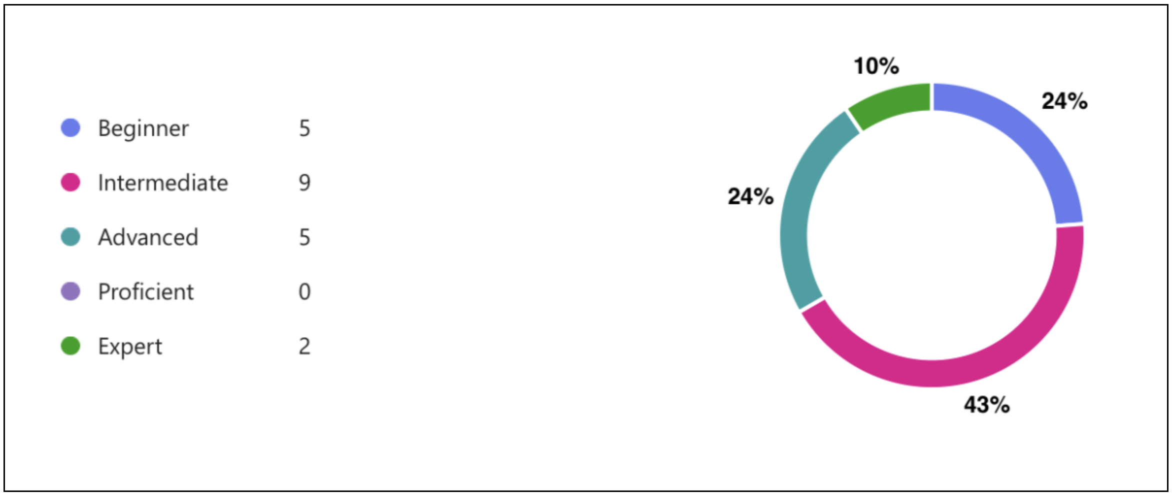


Figure 3.4: Respondents proficiency with data visualization tools

Most respondents consider themselves as intermediate while using data visualization tools. Beginners and advanced are evenly split but are the majority when combined. Two of the respondents are expert in using data visualization tools.

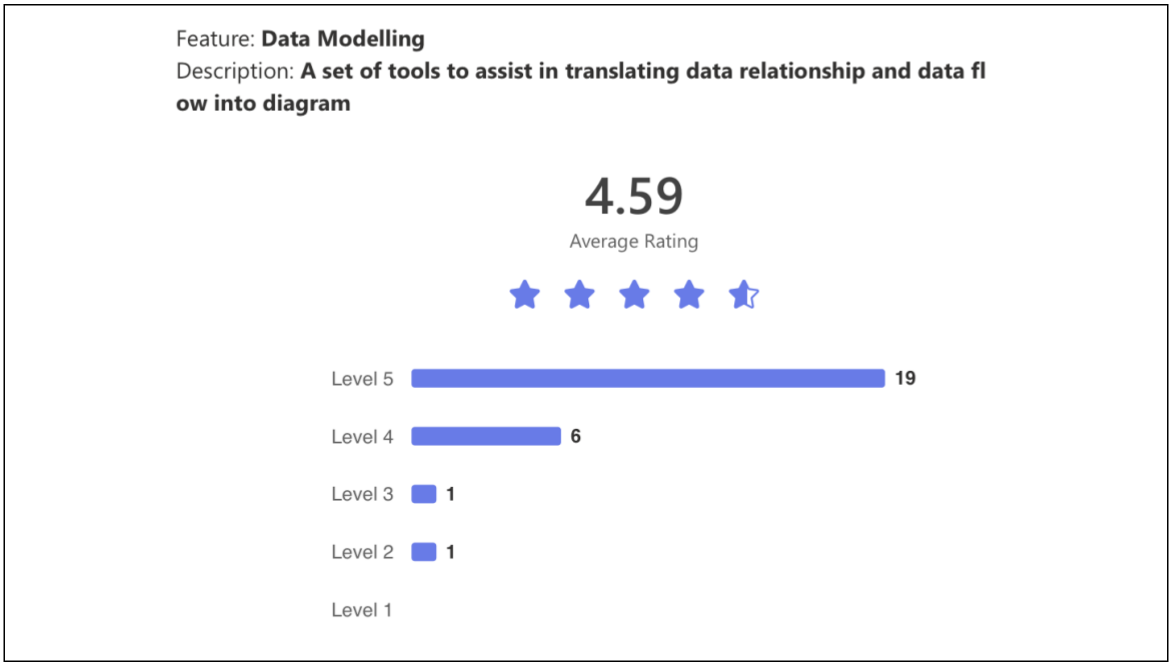


Figure 3.5: Data modelling feature rating

Data Modelling received the highest rating of 4.59 with 19 users rating it at Level 5. This indicates that users find it highly effective for translating data relationships into diagrams. 6 users rated it at Level 4, showing strong satisfaction with minor areas for improvement. A very small number of users rated it at Level 3 and Level 2 which suggests that most find this feature useful and well-implemented.

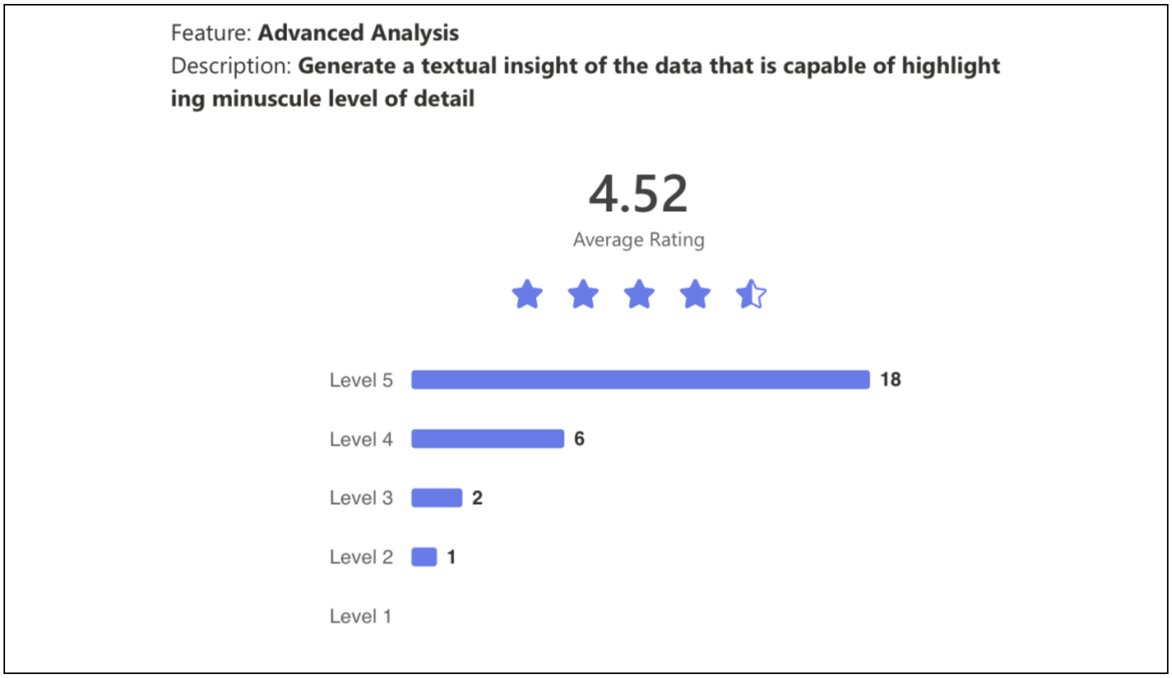


Figure 3.6: Advanced analysis feature rating

Advanced Analysis had an average rating of 4.52 with 18 users rating it at Level 5. This suggests that users appreciate its ability to generate detailed textual insights from data. 6 users rated it at Level 4, indicating a positive experience with slight room for enhancement. 3 users rated it at Level 3 or below.

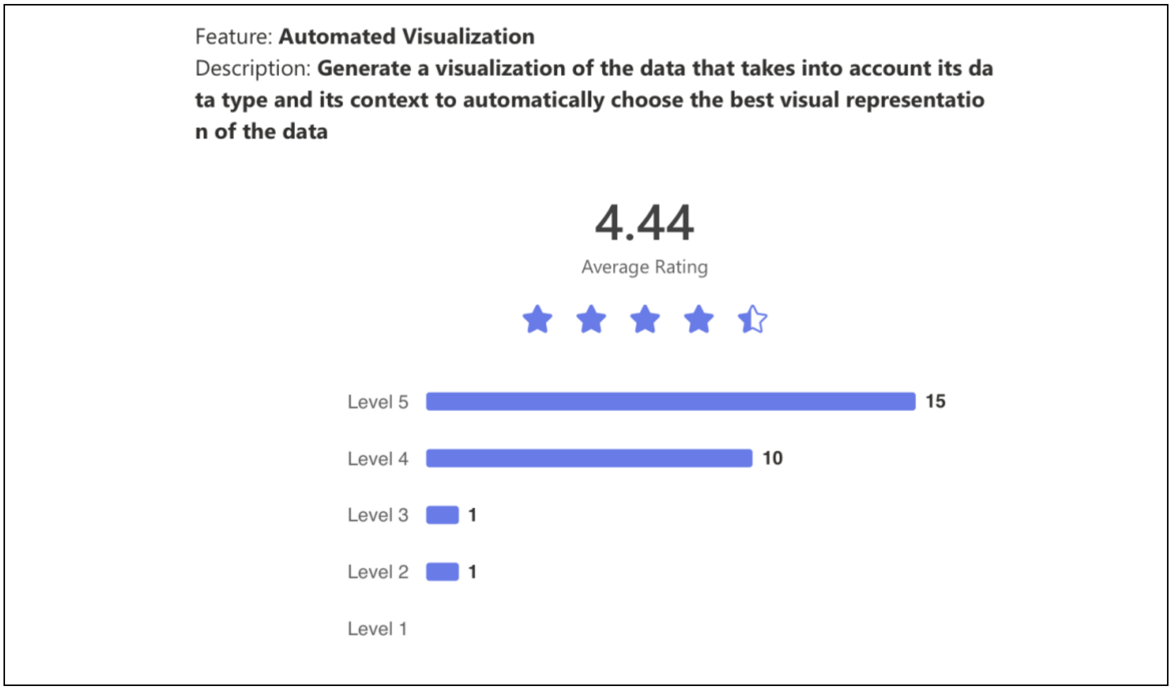


Figure 3.7: Automated visualization feature rating

Automated Visualization received an average rating of 4.44 with 15 users rating it at Level 5. Users find it valuable for selecting the best visual representation of data based on context. 10 users rated it at Level 4, reflecting a generally positive experience. Only 2 users rated it at Level 3 or Level 2 which suggests that the feature performs well for most users with minimal complaints.

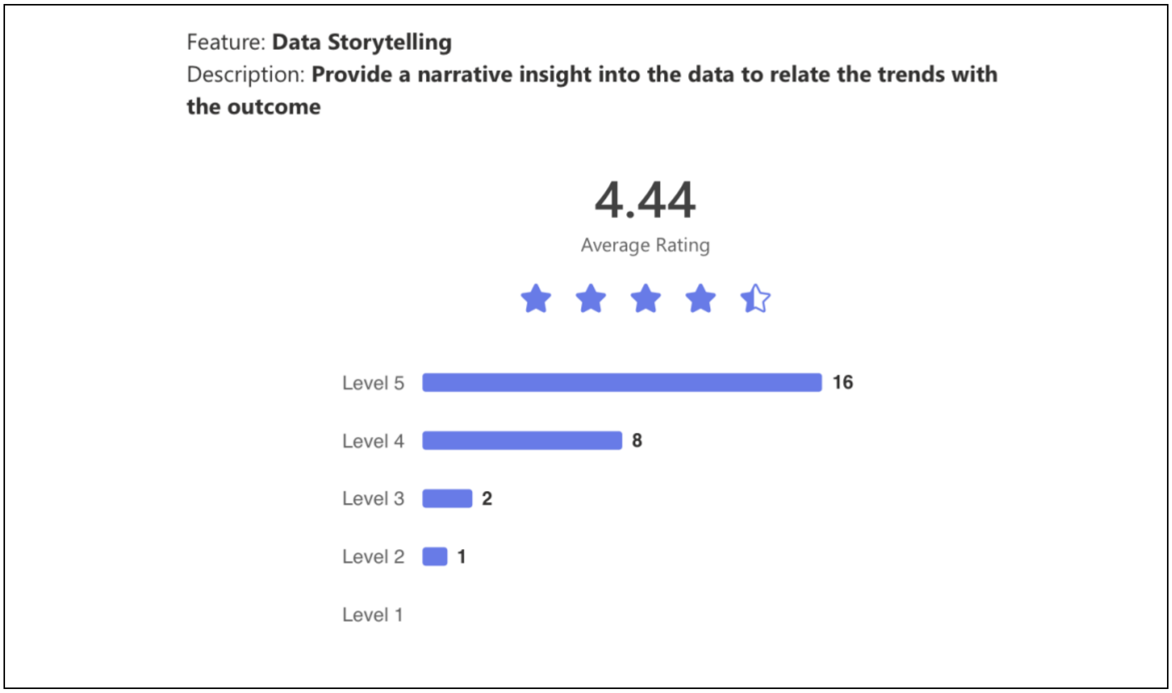


Figure 3.8: Data storytelling feature rating

Same with Automated Visualization, Data Storytelling also had an average rating of 4.44 with 16 users rating it at Level 5. This suggests that many users find it effective in providing narrative insights that relate trends to outcomes. 8 users rated it at Level 4, indicating a positive response with some minor areas for refinement. 3 users rated it at Level 3 or Level 2, showing that a small number of users may not find it as effective for storytelling purposes.

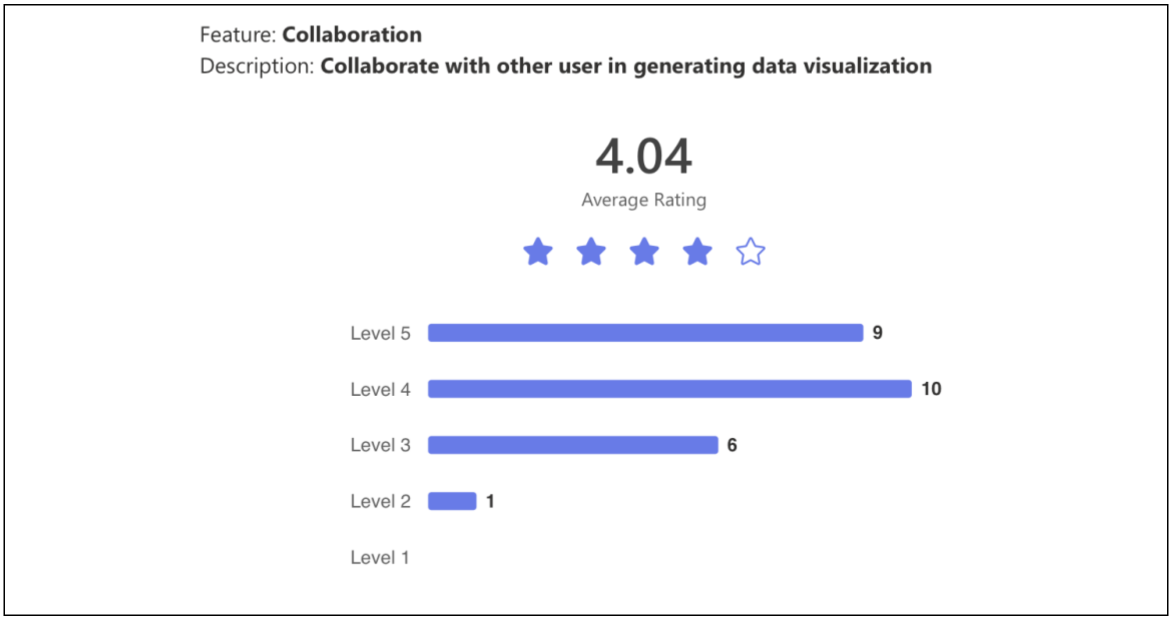


Figure 3.9: Collaboration feature rating

Lastly, Collaboration had the lowest average rating of 4.04 with 9 users rating it at Level 5. 10 users rated it at Level 4 which shows that while many users find it useful, it is not as highly rated as other features. 6 users rated it at Level 3, suggesting that a notable portion of users found it to be average, while 1 user rated it at Level 2.

Table 3.4: Features respondents wish to see in data visualization tools

No.	Responses
1	*so far no*
2	*Good*
3	*Spatial data analysis*
4	*Nope.*
5	*capability to import open source viz, i.e d3 to be integrate in the tool*
6	*Integration of LLM in BI tools to provide insights*
7	*No.*
8	*One feature I love to see in data visualization tools is an AI-powered Insight Generator. Imagine if the tool didn't just display charts but also analyzed your data and highlighted key patterns or trends for us and make it easier way.*
9	*Not sure*
10	*Ai helper*
11	-
12	*auto size the size of column in power bi*
13	*No*
14	*In power bi, i wish to see data cleansing features. Although there is other data cleansing tools, but integrating data cleansing features could improve the efficiency of work.*
15	*1. Adaptive Visualization AI Assistant: A smart, context-aware assistant integrated into visualization tools that dynamically adapts the visualizations based on the dataset, user intent, and target audience. 2. Textual narrative capability using LLMs to interpret and describe charts. This is especially good when working with data simulation when users are able to adjust variables and the the tools reactively regenerate the charts while describing what's happening*
16	*No*
17	*Automatic visualization of tabular data from SQL*

Table 3.5: Limitations on existing data visualization tools

No.	Based on your personal experience, what are the limitations with the existing data visualization tools? Does it affect the goal you are trying to achieve with these tools?
1	*no*
2	*Good*
3	*The dataset itself*
4	*High licensing fees for advanced tools can be a barrier for individuals or small organizations. Budget restrictions may limit access to better tools, forcing compromises on visualization quality.*
5	*limitation number of data points, require certain tweak to viz big data*
6	*Performance bottleneck when dealing with large dataset, since the visualization cannot utilize all data that is available.*
7	*No.*
8	*I think limited interactivity in some tools. While many allow basic filters and drill-downs, they often dont support more dynamic exploration, like asking follow-up questions or visualizing?*
9	*Yes*
10	*limited GB for unuser*
11	-
12	*No*
13	*Limitations in data size and loading time for big data.*
14	*The performance of the existing data visualisation tools depends on the data volume as it tends to slow down when handling massive volumes of data and it can affect the work process.*
15	*Some tools are overly complicated and requires programming skills. Need to be able to tell or select type of charts to draw and the tools should guide users accordingly*
16	*No*
17	*Too many steps*

3.2.3.4. Analysis

The respondent background consists mostly of students and software developers with a smaller number identifying as data analysts and business intelligence professionals. A few respondents fall under other professional categories, including visual editors, customer service representatives, and managers.

The usage frequency of data visualization tools is fairly balanced, with most respondents using them daily, weekly, or monthly. Only one respondent reported using them yearly, and a notable number indicated they never use data visualization tools.

Power BI is the most widely used tool, followed by Tableau. These two are considered the dominant data visualization tools amongst respondents. Looker Studio and Oracle Analytics Cloud are less commonly used which indicates that they are less favored in comparison. Some respondents mentioned other tools, such as Jupyter Notebook, SPSS, Metabase, and Kibana, though Jupyter Notebook would not be considered as a data visualization tool, it is able to generate data visualization with the help of supported libraries and packages.

In terms of proficiency, most respondents consider themselves intermediate users. The number of beginners and advanced users is evenly split, but together they form a significant majority. Only two respondents identified as experts, meaning that high-level proficiency in data visualization tools is less common amongst the surveyed group.

From this group of respondents, the feature ratings indicate that respondents highly value tools that enhance data interpretation and visualization with Data Modelling receiving the highest rating of 4.59. Users find this feature effective in translating data relationships into diagrams with 19 users rating it at Level 5. The small number of lower ratings suggests that most users find it well-implemented with minimal issues.

Advanced Analysis follows closely with a rating of 4.52, showing that users appreciate the ability to generate textual insights from data. The high number of Level 5 ratings indicates that this feature is widely accepted.

Automated Visualization and Data Storytelling both received a rating of 4.44 which indicates that users see them as valuable tools for selecting visual representations and generating narrative insights. The strong number of Level 5 ratings suggests that most users benefit from these features, but some minor areas for improvement remain.

Collaboration received the lowest rating at 4.04 which suggests that while some users find it useful, it does not perform as well as other features. With more users rating it at Level 3 and Level 4, the feedback indicates that collaborative functionalities may need enhancements to be implemented inside of a data visualization tool.

Overall, the analysis shows that features focused on data modeling, analysis, and visualization are highly rated and well-received, while collaborative aspects may require refinement. Users seem to prefer tools that simplify complex data relationships and provide automated insights,

From the open-ended questions, features that respondents wish to see are spatial data analysis, integration of open-source visualization libraries, auto-sizing of columns in Power BI, and built-in data cleansing feature. In addition, some of the features relates closely to the features observed earlier are AI-powered insight generator and textual narrative generation using LLM, which is similar to the advanced analysis feature from the questionnaire and observation. In addition, the automated visualization feature is also mentioned in the open-ended question responses where respondents wish to see automatic visualization from SQL tables and adaptive visualization AI assistant. Lastly, respondents also wish to see LLM integrated in BI tools, which relates very closely to the solution proposed in this project.

From the limitations of existing data visualization tools, user struggles with high licensing costs, handling large datasets, efficient data processing, enhanced creativity, storage limitations, and improved data loading speed. The limitations of simplified user experience and reduction steps for visualization relates closely to the project goal which could possibly be solved through this proposed data visualization tool.

From here, the rated features will be mapped against the result from the observation and the open-ended questions with the scope and the goal of the project in mind.

3.2.4. Requirement Analysis Summary

After mapping the result from the observation and analysis, it is concluded that data connection, chart & visualization, dashboard, and advanced analysis are highly prominent in defining a data visualization tool based on the observation and questionnaire closed-ended and open-ended response. They are a combination of a must in every data visualization tools and user sees it as very useful and important in order to execute data visualization tasks. In addition, some of these features are features that was mentioned by the questionnaire respondent under the 'features they wish to see' question.

The features that were discarded are deemed not important enough as it does not exist in every data visualization tool and the responses from the questionnaire are not enough to justify the importance of its presence in a data visualization tool.

3.3. Functional Requirements

Table 3.6 displays the finalized functional requirements of the system after going through the requirements elicitation process. These functions are expected to be available in the system and are expected to solve the problems that exists within existing data visualization tools.

Table 3.6: System’s functional requirements

Requirement
The user shall be able to create an account by registering their credentials into the system
The system shall store user credentials during registration
The user shall be able to login to the system using their credentials
The system shall validate user credentials when logging in
The user shall be able to enter the data source credentials into the system
The system shall connect to the data source provided by the user
The system shall retrieve the database schema from the data source provided by the user
The user shall be able to submit a prompt into the system
The LLM shall generate executable SQL query
The system shall extract data from the data source and store the data temporarily within the system
The LLM shall convert the extracted data from the data source into a structured format
The LLM shall choose and produce visualization of the data to be displayed on the interface
The LLM shall generate analysis of the data to be displayed on the interface
User shall be able to choose the visuals for the LLM to consider when choosing a visual for the data
User shall be able to save the generated visual and analysis from the LLM

Table 3.7 to table 3.16 displays the use cases of the system and its details of actors, description, precondition, postcondition, and its scenarios.

Table 3.7: Register use case description

Requirement ID	REQ_F001	**Requirement Name**	Register
Actor / User	User		
Description	User enters their email, username, and password into the system to create an account and access the system		
Precondition	User has not created an account yet		
Postcondition	User credentials are stored within the system		
Main Success Scenario	1. User input their email, username, and password 2. The inputs are validated by the system 3. The inputs are stored as in the system 4. The system displays the success message		
Alternative Scenario	1. User inputs their email, username, and password 2. The inputs are validated by the system 3. The account credentials already exist in the system 4. The system displays the alert message		
Exception Scenario	1. User inputs their email, username, and password 2. The inputs are invalidated by the system 3. The system displays the error message		
Author	Luqman Hakim bin Noorazmi		

Table 3.8: Login use case description

Requirement ID	REQ_F002	**Requirement Name**	Login
Actor / User	User		
Description	User enters their username or email and password into the system to access the system		
Precondition	User has registered to the system		
Postcondition	User has access to the system using their profile		
Main Success Scenario	1. User inputs their email or username and password 2. The inputs are validated by the system 3. The system displays the success message 4. User are given access to the system		
Alternative Scenario	-		
Exception Scenario	1. User inputs their email or username and password 2. The inputs are invalidated by the system 3. The system displays the error message		
Author	Luqman Hakim bin Noorazmi		

Table 3.9: Connect to data source use case description

Requirement ID	REQ_F003	**Requirement Name**	Connect to data source
Actor / User	User		
Description	User enters database connection credentials: hostname, username, password, and database name and connect to the data source		
Precondition	The user has access to the system		
Postcondition	The database credentials are stored in the system and the system is connected to the data source		
Main Success Scenario	1. User inputs the database hostname, username, password, and database name. 2. The inputs are validated by the system 3. The inputs are stored as database credentials in the system. 4. The system attempts to connect to the data source 5. The connection attempt successful 6. The system displays the success message.		
Alternative Scenario	-		
Exception Scenario	1. User inputs the database hostname, username, password, and database name. 2. The inputs are validated by the system 3. The inputs are stored as database credentials in the system. 4. The system attempts to connect to the data source 5. The connection attempt unsuccessful 6. The system displays the error message.		
Author	Luqman Hakim bin Noorazmi		

Table 3.10: Submit prompt use case description

Requirement ID	REQ_F004	**Requirement Name**	Submit Prompt
Actor / User	User		
Description	The user enters prompt into the system		
Precondition	The user has access to the system The connection to the data source is established		
Postcondition	The prompt is stored in the system		
Main Success Scenario	1. The user enters the prompt 2. The prompt is validated by the system 3. The prompt is stored in the system		
Alternative Scenario	-		
Exception Scenario	1. The user enters the prompt 2. The prompt is invalidated by the system 3. The system displays the failure message 4. The system displays the error log		
Author	Luqman Hakim bin Noorazmi		

Table 3.11: Select visuals use case description

Requirement ID	REQ_F005	**Requirement Name**	Select Visuals
Actor / User	User		
Description	User selects the visuals for the LLM to consider when generating visualization		
Precondition	The user has access to the system		
Postcondition	The selected tools are ready to be passed to the LLM		
Main Success Scenario	1. User selects the visual from the available visual choices 2. The selections are updated within the system 3. The system displays the success message.		
Alternative Scenario	-		
Exception Scenario	-		
Author	Luqman Hakim bin Noorazmi		

Table 3.12: Generate SQL query use case description

Requirement ID	REQ_F006	**Requirement Name**	Generate SQL Query
Actor / User	LLM		
Description	The LLM translates the prompt by the user into an executable SQL query		
Precondition	The user has access to the system The connection to the database is established The database schema is fetched		
Postcondition	The prompt by the user is translated into an executable SQL query		
Main Success Scenario	1. The system gets the stored database schema and prompt 2. The LLM generates the SQL with consideration to the database schema and prompt 3. The SQL query is validated 4. The SQL query is stored in the system 5. The system displays the success message		
Alternative Scenario	1. The system gets the stored database schema and prompt 2. The LLM generates the SQL with consideration to the database schema and prompt 3. The SQL query is invalidated The SQL query is generated again with consideration to the previous SQL query		
Exception Scenario	1. The system gets the stored database schema and prompt 2. The LLM generates the SQL with consideration to the database schema and prompt 3. The SQL query is invalidated 4. The SQL query is generated again with consideration to the previous SQL query 5. The LLM reaches the maximum attempt for SQL query regeneration 6. The system displays the failure message		
Author	Luqman Hakim bin Noorazmi		

Table 3.13: Generate data dictionary use case description

Requirement ID	REQ_F007	**Requirement Name**	Generate Data Dictionary
Actor / User	LLM		
Description	LLM converts the queried data from the data source into a structured format		
Precondition	The data has been queried from the data source		
Postcondition	The data are transformed into a dictionary		
Main Success Scenario	1. The system gets the stored data from data source 2. LLM generate dictionary based on the queried data 3. The dictionary is validated by the system 4. The dictionary is stored in the system 5. The system displays the success message		
Alternative Scenario	-		
Exception Scenario	4. The system gets the stored data from data source 5. LLM generate dictionary based on the queried data 6. The dictionary is invalidated by the system 7. The system displays the failure message		
Author	Luqman Hakim bin Noorazmi		

Table 3.14: Generate visual use case description

Requirement ID	REQ_F008	**Requirement Name**	Generate Visual
Actor / User	LLM		
Description	LLM generates the best suited visual for the data		
Precondition	The prompt has been entered The data dictionary has been generated		
Postcondition	The visual representation of the data is generated		
Main Success Scenario	1. The system gets the data dictionary and prompt 2. The LLM evaluates the visual selection 3. The LLM chooses the visual 4. The visual is stored in the system 5. The visual is displayed		
Alternative Scenario	-		
Exception Scenario	1. The system gets the data dictionary and prompt 2. The LLM evaluates the visual selection 3. The LLM is unable to chooses the visual 4. The system displays the failure message		
Author	Luqman Hakim bin Noorazmi		

Table 3.15: Generate analysis use case description

Requirement ID	REQ_F009	**Requirement Name**	Generate Analysis
Actor / User	LLM		
Description	The LLM generates a textual analysis of the data		
Precondition	The prompt has been entered The data has been queried from the data source		
Postcondition	The analysis of the data is generated		
Main Success Scenario	1. The system gets the retrieved data and prompt 2. The LLM generates the analysis of the data 3. The system displays the analysis of the data		
Alternative Scenario	-		
Exception Scenario	1. The system gets the retrieved data and prompt 2. The LLM is unable to generate the analysis of the data 3. The system displays the failure message		
Author	Luqman Hakim bin Noorazmi		

Table 3.16: Save visual and analysis use case description

Requirement ID	REQ_F010	**Requirement Name**	Save Visual and Analysis
Actor / User	User		
Description	User saves the generated visual and analysis		
Precondition	The visual and analysis are displayed on the user interface		
Postcondition	The visual and analysis instance is stored within the system based on the user ID		
Main Success Scenario	1. The system displays the visual and analysis 2. User choose the save visual and analysis option 3. Visual and analysis are stored within the system 4. The system displays the success message		
Alternative Scenario	-		
Exception Scenario	1. The system displays the visual and analysis 2. User choose the save visual and analysis option 3. Visual and analysis fail to be stored within the system 4. The system displays the failure message		
Author	Luqman Hakim bin Noorazmi		

3.4. Quality Requirements

For every process that involves LLM, the quality of the response is important. The quality requirements are considered based on the current resources' quality. Table 3.16 to table 3.23 describes the quality requirements of the system through several different metrices of performance, portability, and usability (Krüger, 2024).

To evaluate the performance metric, the LLM's response time can be used to measure the model's response time based on the unit of text input, or token, to the model. For rough estimation purpose, the expected input for each of LLM's use case are defined.

Based on the Llama 3.3 70B Versatile model's performance on Groq hardware we can estimate execution times for key operations. Generating an

SQL query takes approximately 0.36 seconds, as it is expected to process around 100 tokens at 276 tokens per second. Creating a data dictionary requires about 150 tokens, leading to an estimated execution time of 0.54 seconds. Selecting the best visualizationinvolves approximately 200 tokens, making the process take 0.72 seconds. Finally, generating an in-depth analysis using also processes around 150 tokens, resulting in an execution time of 0.54 seconds. These estimates are based on the reported inference speed of 276 tokens per second on Groq hardware (Groq, Inc., n. d.), though actual times may vary depending on input complexity and system performance.

Table 3.17: Visualization per minute quality description

Requirement ID	REQ_Q001	**Requirement Metric**	Performance
Requirement Sub-Metric	Visualization per minute		
Description	Amount of visualization request per minute.		
Requirement	Based on the open-source model to be integrated, the Rate Per Minute (RPM) is 30 requests per minute (Groq, Inc., n. d.).		
Author	Luqman Hakim bin Noorazmi		

Table 3.18: SQL Generation response time quality description

Requirement ID	REQ_Q002	**Requirement Metric**	Performance
Requirement Sub-Metric	SQL Generation Response Time		
Description	Amount of time taken to generate SQL query		
Requirement	Approximately 0.36 seconds for around 100 tokens		
Author	Luqman Hakim bin Noorazmi		

Table 3.19: Amount of time taken to generate data dictionary quality description

Requirement ID	REQ_Q003	**Requirement Metric**	Performance
Requirement Sub-Metric	Generate Data Dictionary Response Time		
Description	Amount of time taken to generate data dictionary		
Requirement	Approximately 0.54 seconds for around 150 tokens		
Author	Luqman Hakim bin Noorazmi		

Table 3.20: Amount of time taken to generate visual quality description

Requirement ID	REQ_Q004	**Requirement Metric**	Performance
Requirement Sub-Metric	Generate Visual Response Time		
Description	Amount of time taken to generate visual		
Requirement	Approximately 0.72 Seconds for around 200 tokens		
Author	Luqman Hakim bin Noorazmi		

Table 3.21: Amount of time taken to generate analysis quality description

Requirement ID	REQ_Q005	**Requirement Metric**	Performance
Requirement Sub-Metric	Generate Analysis Response Time		
Description	Amount of time taken to generate analysis		
Requirement	Approximately 0.54 seconds for 150 tokens		
Author	Luqman Hakim bin Noorazmi		

Table 3.22: Time for user to familiarize with the system quality description

Requirement ID	REQ_Q006	**Requirement Metric**	Usability
Requirement Sub-Metric	Training Time		
Description	Amount of time for user to familiarize with the system on the first interaction.		
Requirement	Without considering the registration and login processes, time between setting up the data source and generating visualization is approximately 10 minutes. The simple interface and familiarity with existing AI-driven system that accepts prompt could heavily influence this metric.		
Author	Luqman Hakim bin Noorazmi		

Table 3.23: Percentage portable code quality description

Requirement ID	REQ_Q007	**Requirement Metric**	Portability
Requirement Sub-Metric	Percentage of non-portable code		
Description	Percentage of the code that cannot be converted into different platform (web app, mobile phone, etc.).		
Requirement	75% of the code should be non-portable due to the implementation of Python-specific library for the interface, PyQt5.		
Author	Luqman Hakim bin Noorazmi		

Table 3.24: Number of operating systems supported quality description

Requirement ID	REQ_Q008	**Requirement Metric**	Performance
Requirement Sub-Metric	Number of systems where software can run.		
Description	Number of operating systems the proposed system can be functional.		
Requirement	Three (3) operating systems will be capable of running the system. The operating systems are Windows, Linux, and Mac.		
Author	Luqman Hakim bin Noorazmi		

3.5. System Context Diagram

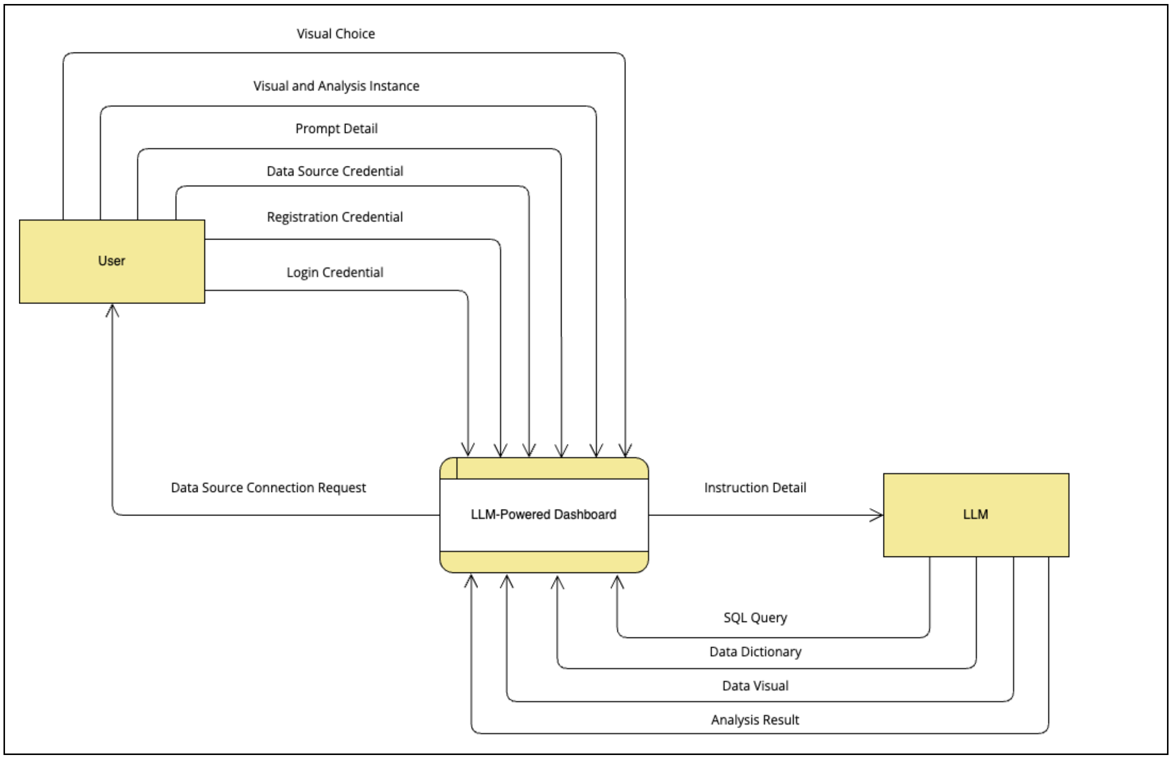


Figure 3.10: LLM-Powered Dashboard Context Diagram (Anisya et al., 2021)

Figure 3.10 depicts the context diagram of the proposed data visualization tool. A context diagram provides a high-level overview of the data visualization tool by defining its boundaries and interactions with external entities through data that flow between them. It serves as a tool for global system design that illustrate the data visualization tool and its components in a broad and generalized manner (Anisya et al., 2021).

The external entities are user which also acts as the data source provider, and the LLM. User interacts with the system through the visual choice, visual and analysis instance, prompt detail, data source credential, registration credential, and login credential. The system's interaction with the user is through the data source connection request, while the system's interaction with the LLM is through the instruction detail. The LLM is capable of interacting with the system through the generated SQL query, data dictionary, data visual, and data analysis.

3.6. Data Analysis Plan

Data storage is required for the system to store user's credentials and preferences. A lightweight and easy-to-use database is sufficient to store user's credentials and setting preferences locally. After some consideration, it was found that for Windows, macOS, and Linux, SQLite is a straightforward and portable database that may be used to store user data locally. Python is one of the numerous programming languages that it supports by default. SQLite also makes data management and use simple by storing it in a single file.

The first step to designing the data is to paint the data structure visually. A visual model used to depict corporate entities, their characteristics, and their relationships is called an entity-relationship diagram (ERD). Database design begins with the creation of an ER model. After it is finished, tables with primary and foreign keys are produced by using translation rules to transform it into a logical data model. In order to cut down on redundancy and boost efficiency, the model is then examined for normalization (Rashkovits & Lavy, 2021). Figure 3.11 displays the ERD of the system.

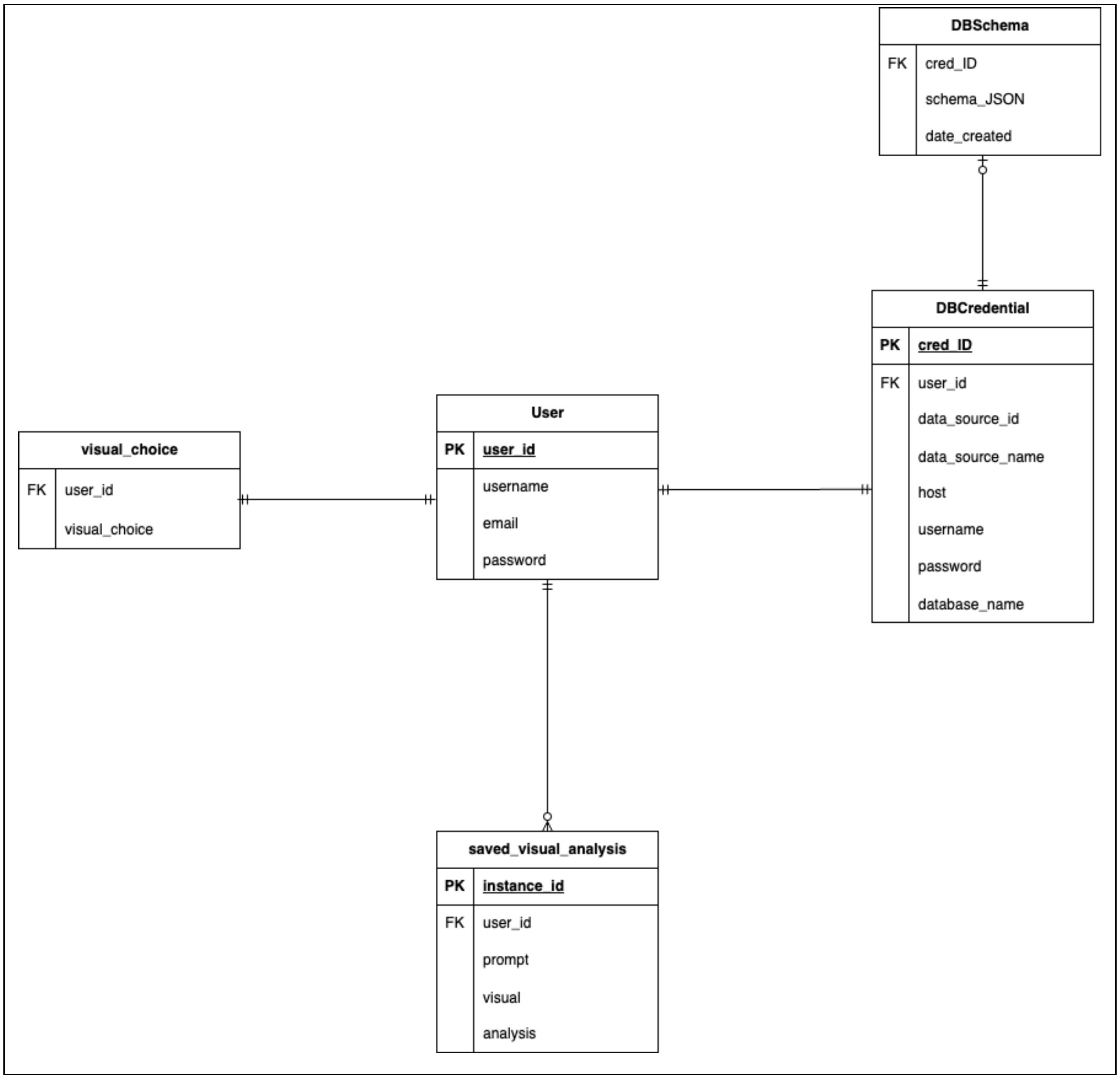


Figure 3.11: System's entity relationship diagram (Rashkovits & Lavy, 2021)

User's preferences such as their visual choices and saved visual and analysis will be stored through the *visual_choice* and *saved_visual_analysis* table. User's data source credentials will be stored in the *DBCredential* table with its own unique ID to ensure the instance in the *DBSchema* table is always up to date with the latest data source credentials.

User has the option to save any visual and analysis. In addition, each user must have their own visuals choices and data source credentials. Table 3.16 depicts the data dictionary of the system based on the presented ERD.

Table 3.25: System's data dictionary

Field Name	Data Type	Data Format	Field Size	Description
user_id	TEXT	XNN-NNX-NXX	11	Unique identifier for each user
username	TEXT	XNNNNNNN	8	User's chosen username
email	TEXT	XXX@XXX.com	50	User's email address
password	TEXT	XNNNNNNNNNN	11	Encrypted password for authentication
visual_choice	JSON	{X:X}	1000	User's selected visualization option JSON
instance_id	TEXT	XNN-NNX-NXX	11	Unique identifier for each saved analysis
prompt	TEXT	XNNNNNNNNNNNNNNNN	17	User's input query for visualization
visual	TEXT	XNNNNNNNNNNNNNN	100	Generated visual representation URL
analysis	TEXT	XNNNNNNNNNNNNNN	100	Interpretation of the visualized data URL
cred_ID	TEXT	XNN-NNX-NXX	11	Unique identifier for database credentials
data_source_id	TEXT	XNN-NNX-NXX	11	Unique identifier for the data source
data_source_name	TEXT	XNNNNNNNNNNN	12	Name of the data source
host	TEXT	XNNNNNNNNNN	11	Database host address
username	TEXT	XNNNNNNNN	9	Username for database access
password	TEXT	XNNNNNNNNNNNN	13	Encrypted database password
database_name	TEXT	XNNNNNNNNNNN	12	Name of the database
schema_JSON	JSON	{X:X}	2000	Stored JSON representation of the database schema
date_created	TEXT	YYYY-MM-DD HH:MM:SS	19	Timestamp when the schema was stored

Chapter 4: System Design

4.1. Use Case Diagram

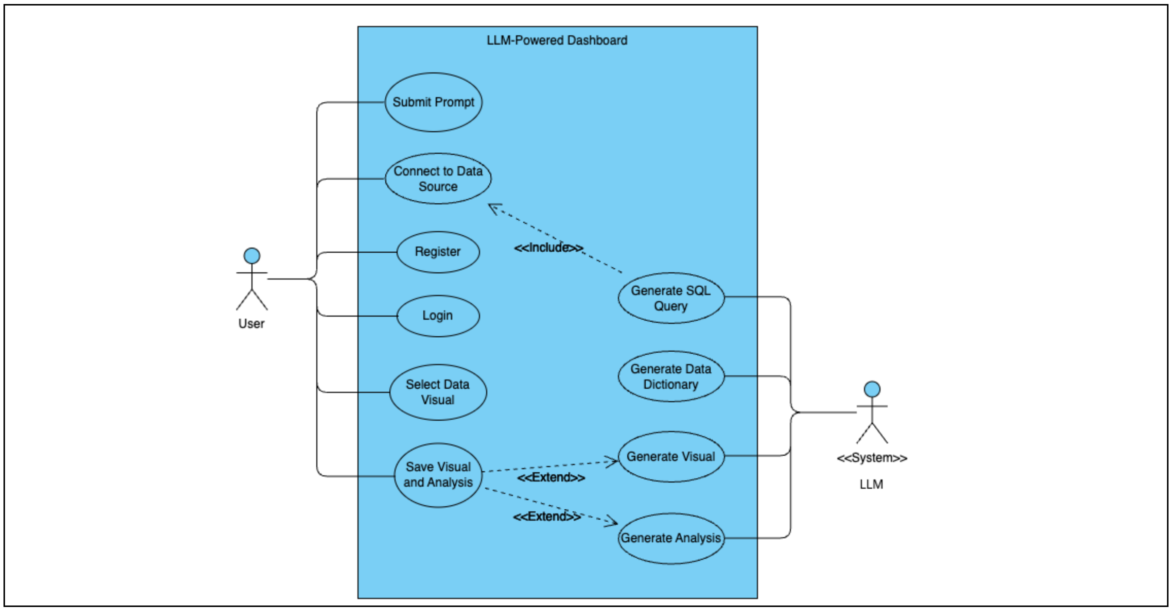


Figure 4.1: System's use case diagram (Sudiartha et al., 2021)

Figure 4.1 displays the use cases of the systems. A use case represents the interaction between a user (actor) and a system. Use case diagrams illustrate a series of interconnected interactions between actors and the system, detailing the types of interactions a user can perform within a program. These diagrams help visualize the functionalities and features available in the system by mapping out how users engage with it (Sudiartha et al., 2021).

The user will be capable of login and register to the system, save the visual and analysis, and entering the data source credentials and the prompt for visualizing data into the system. The database credentials will be used by the system to make connection to the data source. Using the database schema from the data source, the LLM will generate SQL query to extract the data from the data source using the generated query and converts the data into a data dictionary. Lastly, the LLM will generate visual and analysis of the data.

4.2. Sequence Diagram

Micskei and Waeselynck argue that Sequence Diagrams (SDs) serve multiple purposes, such as illustrating the flow of method calls within a program or providing a partial specification of interactions in distributed systems (Al-Fedaghi, 2021). Hamza and Hammad state that SDs can also be used for automating test case generation, making them valuable in software validation processes (Al-Fedaghi, 2021). Lu and Kim explain that SDs depict how events or activities in a use case correspond to operations in class diagrams. They describe events as fundamental behavioral constructs that can be combined into larger behavioral fragments to represent system interactions (Al-Fedaghi, 2021).

For this data visualization tool, the sequence follows the Model View Controller (MVC) architecture. The output from the LLM is encapsulated within the functions from the controllers. The responses are stored within the models and the response are displayed to the user through the interface where it is the only component user will be interacting with the system.

4.2.1. Submit Prompt

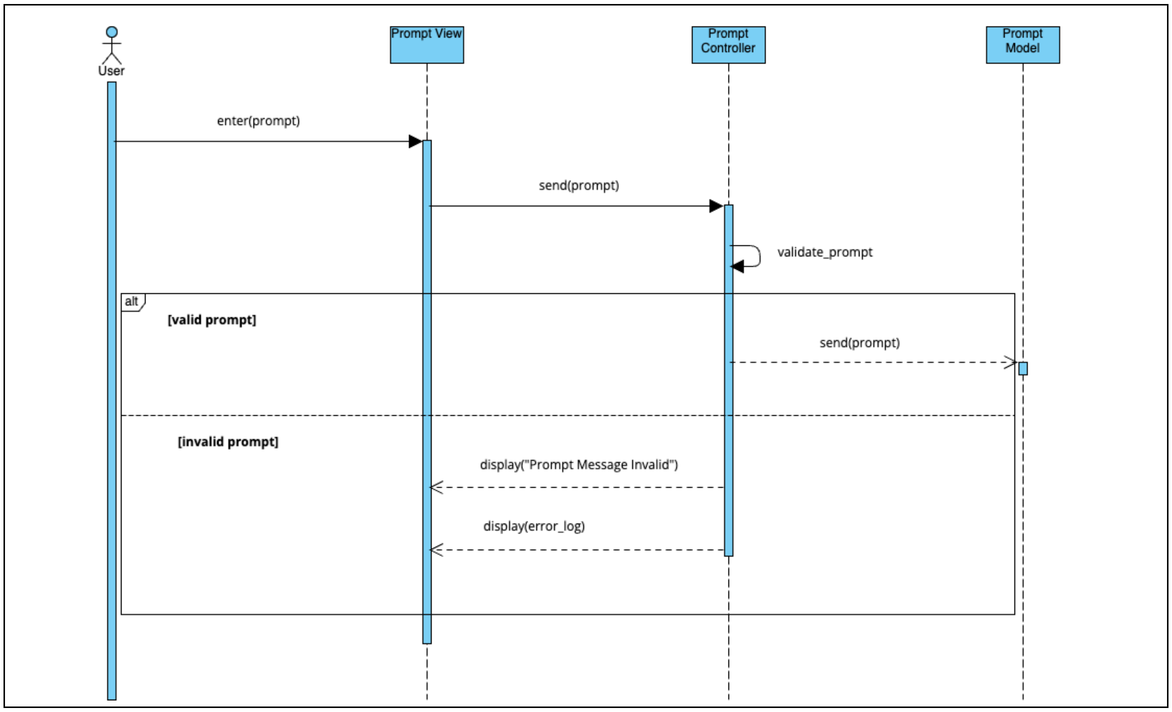


Figure 4.2: Submit prompt sequence diagram (Al-Fedaghi, 2021)

4.2.2. Connect to Data Source

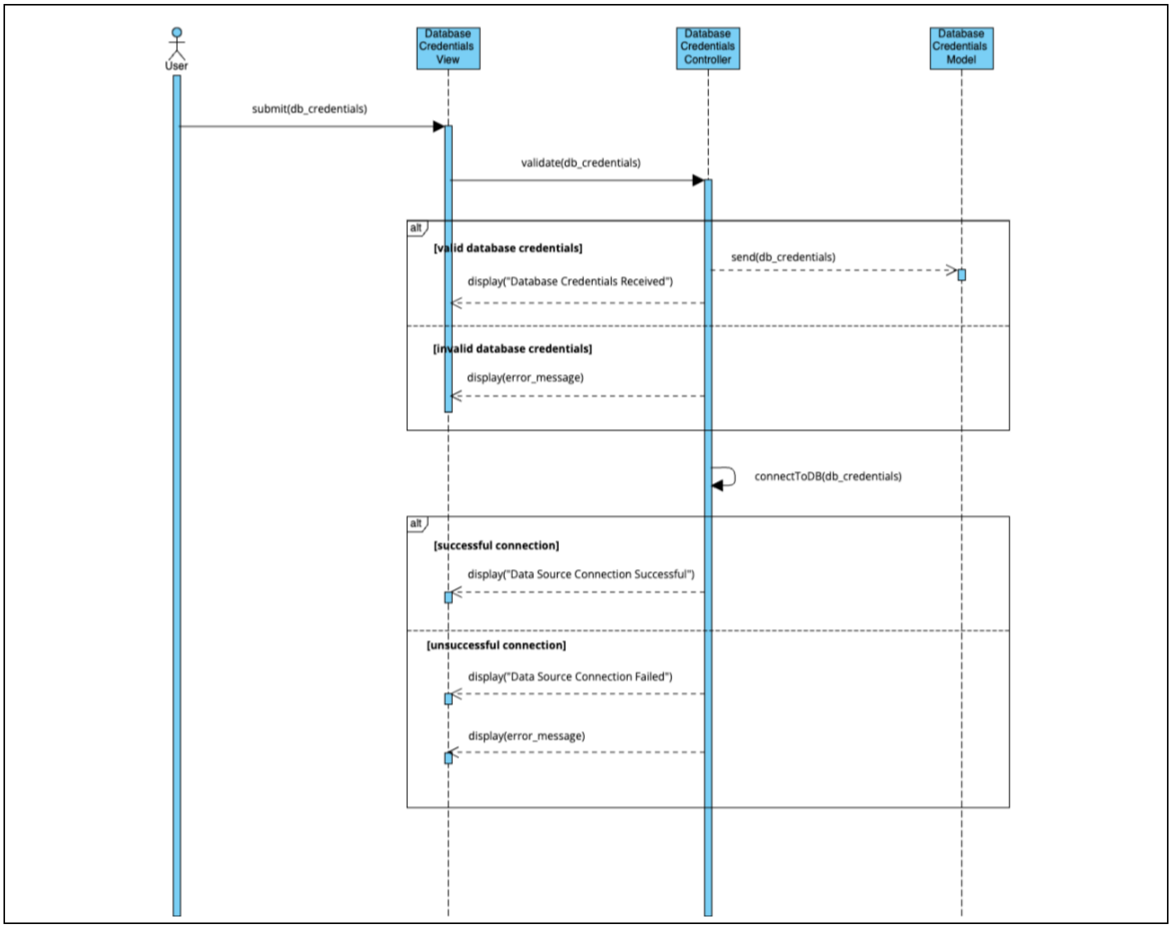


Figure 4.3: Connect to data source sequence diagram (Al-Fedaghi, 2021)

4.2.3. Select Visuals

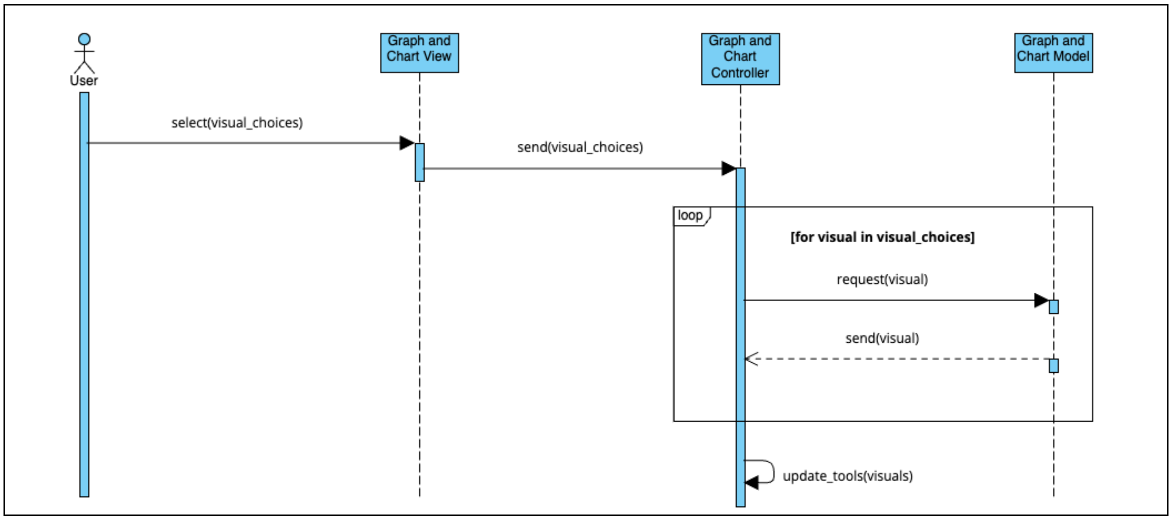


Figure 4.4: Select visuals sequence diagram (Al-Fedaghi, 2021).

4.2.4. Save Visual and Analysis

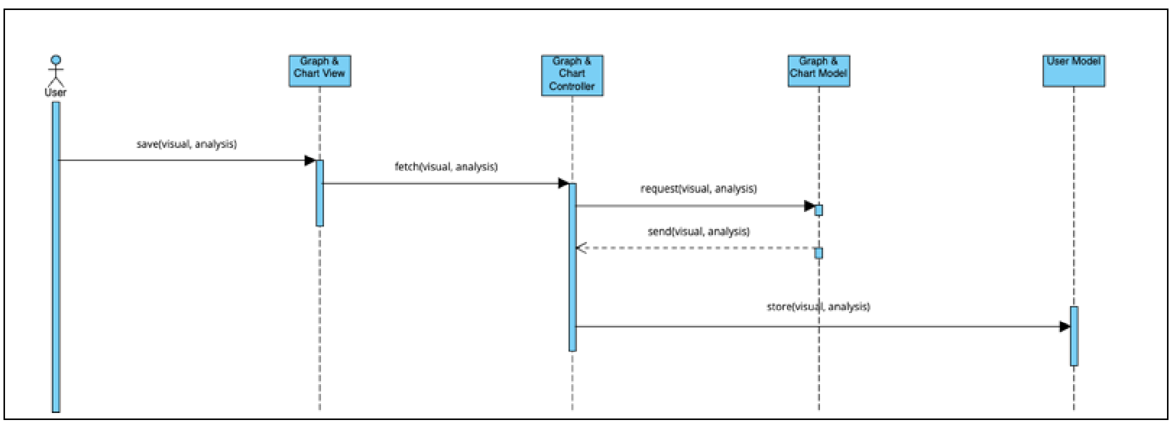


Figure 4.5: Save visual and analysis (Al-Fedaghi, 2021).

4.2.5. Generate SQL

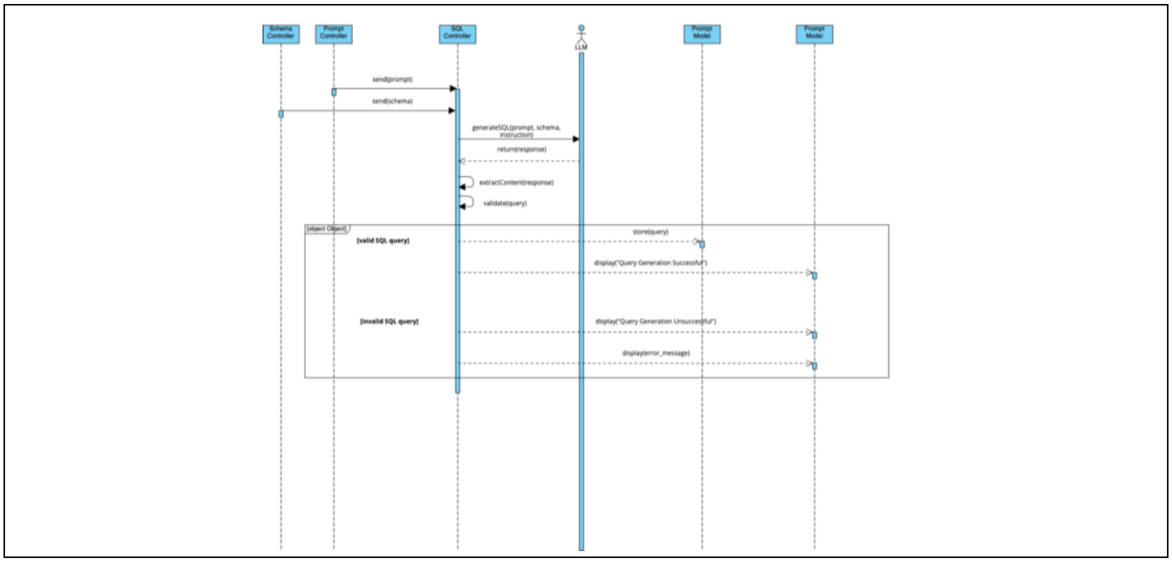


Figure 4.6: Generate SQL sequence diagram (Al-Fedaghi, 2021)

4.2.6. Generate Data Dictionary

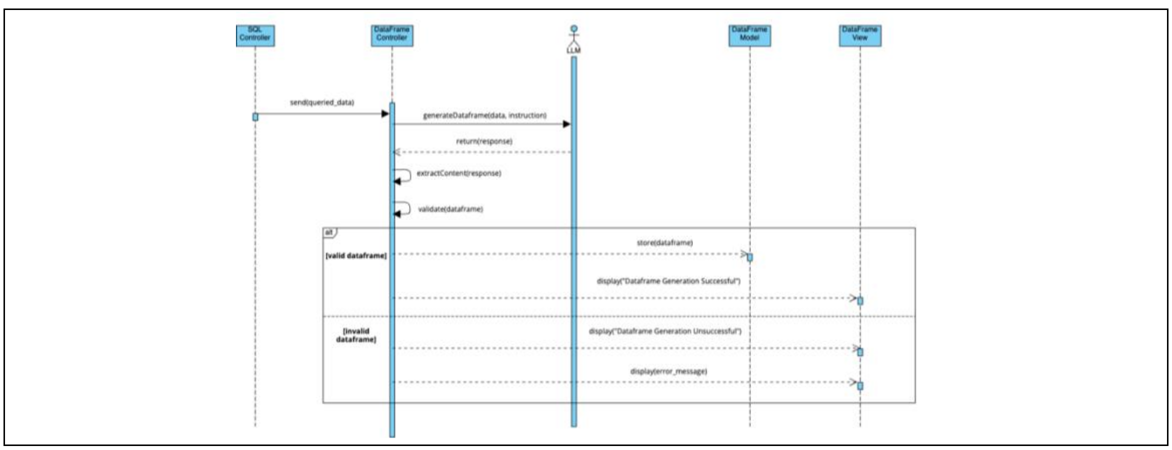


Figure 4.7: Generate data dictionary sequence diagram (Al-Fedaghi, 2021)

4.2.7. Generate Visual

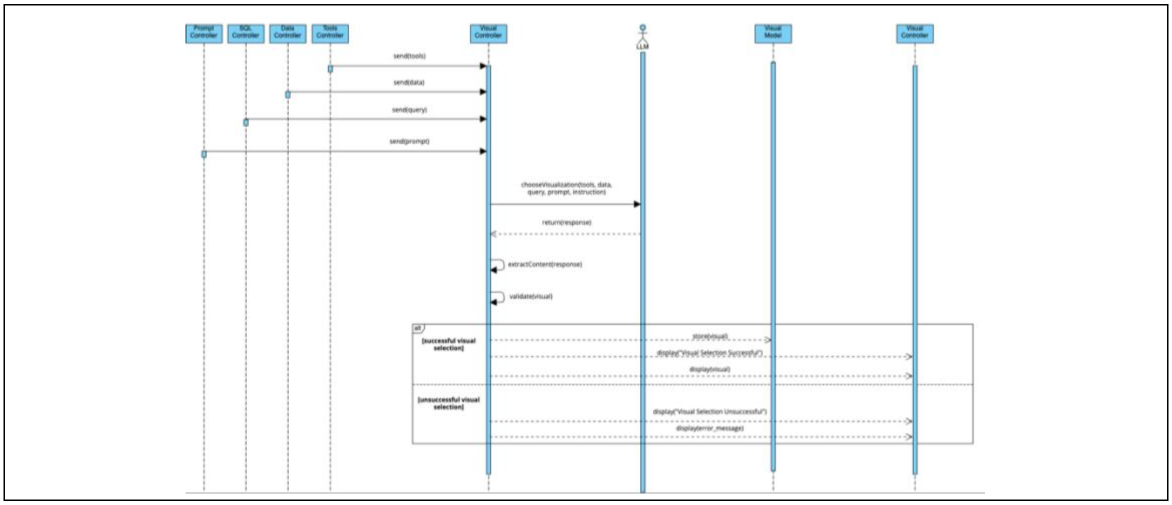


Figure 4.8: Generate visual sequence diagram (Al-Fedaghi, 2021).

4.2.8. Generate Analysis

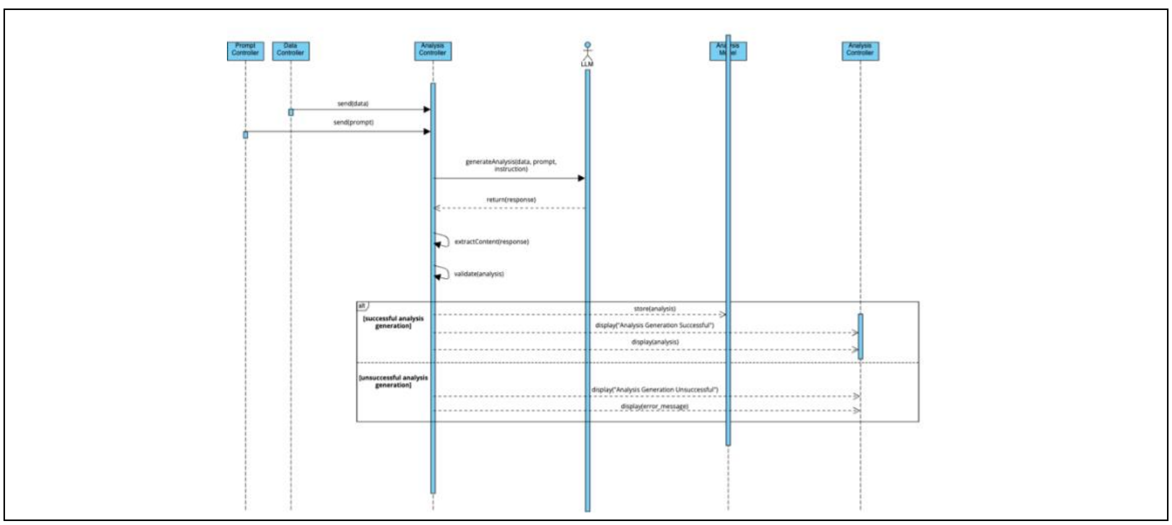


Figure 4.9: Generate analysis sequence diagram (Al-Fedaghi, 2021)

4.3. System Interface

Figure 4.10 to figure 4.15 shows the system's interface concept. The background color is a deep, muted blue-gray. It creates a calm and professional look. This color contrasts with the white text to improve readability. The design stays modern and clean.

The color palette follows a minimalist style. White buttons give a neutral and polished look. Black button text ensures clear legibility. The hover effect adds a slightly richer shade for interactive elements. The background provides a stable foundation to enhance usability and focus. The dashboard will appear structured, user-friendly, and visually appealing.

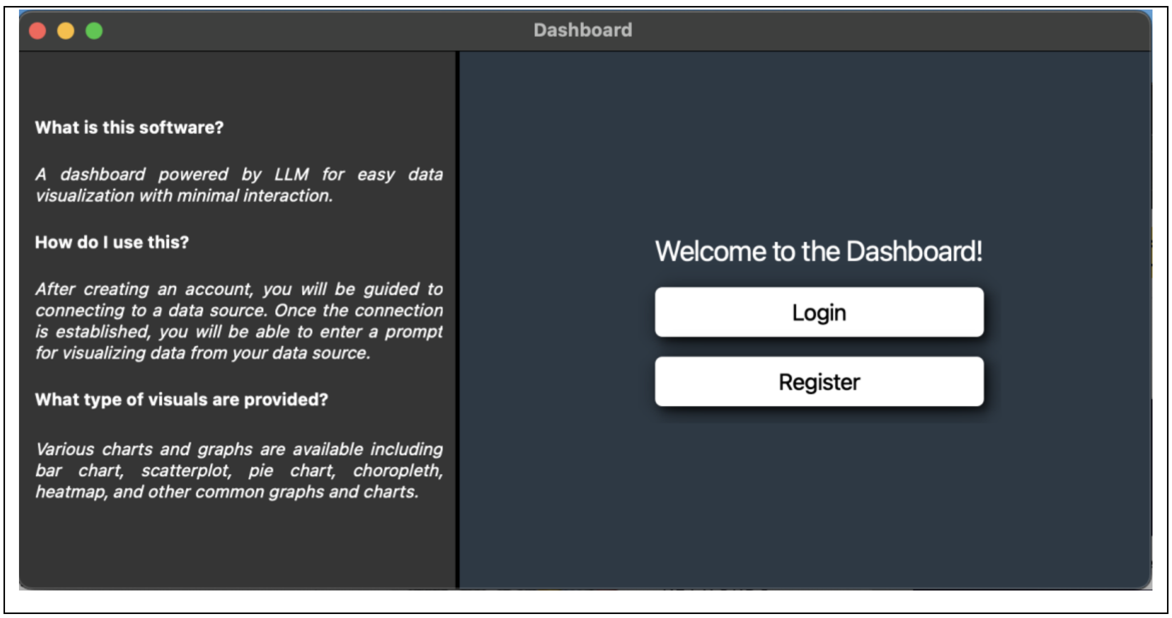


Figure 4.10: Welcome screen

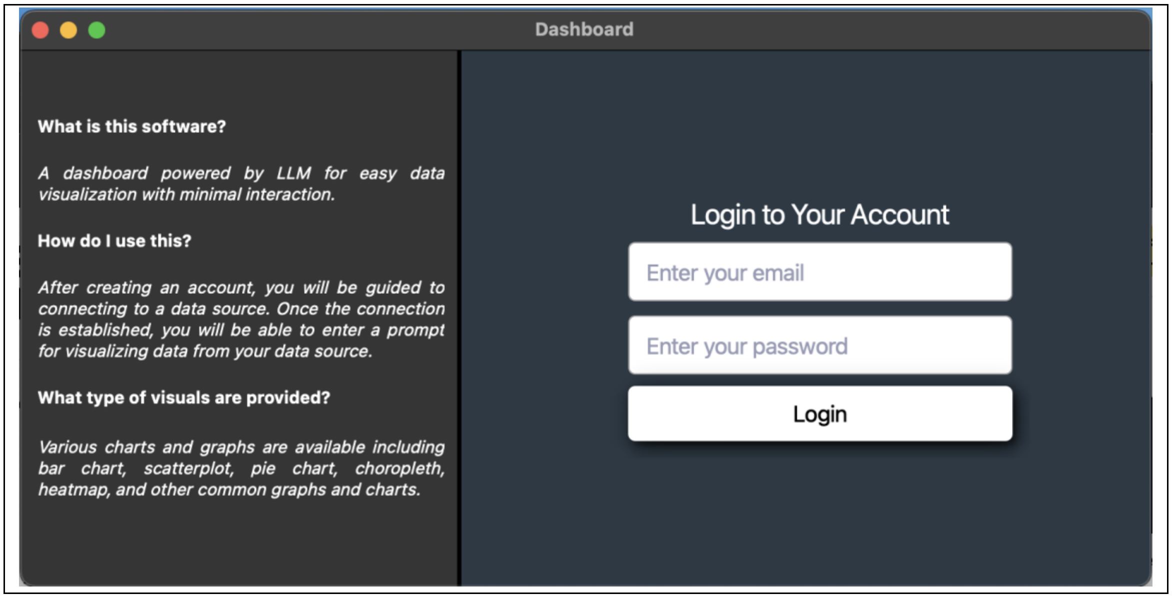


Figure 4.11: Login screen

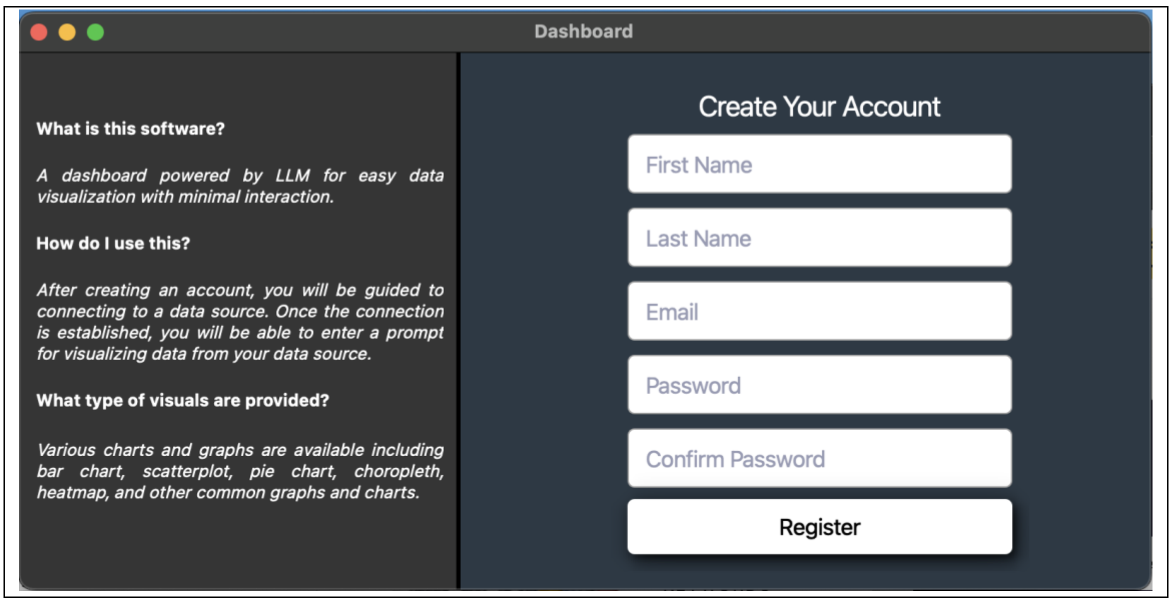


Figure 4.12: Registration Screen

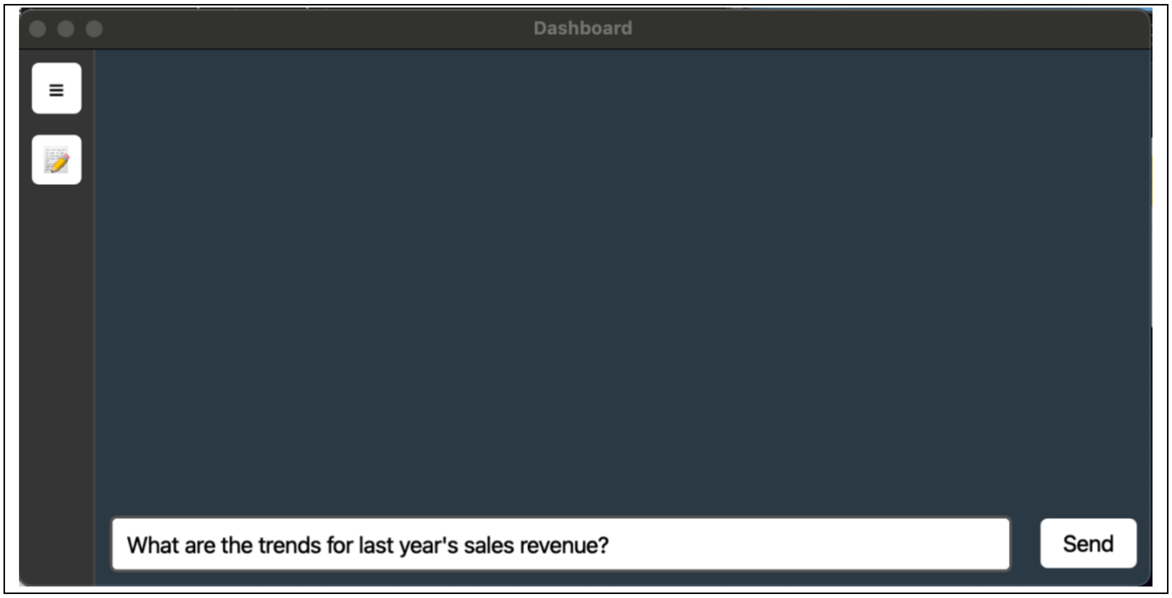


Figure 4.13: Prompt field screen

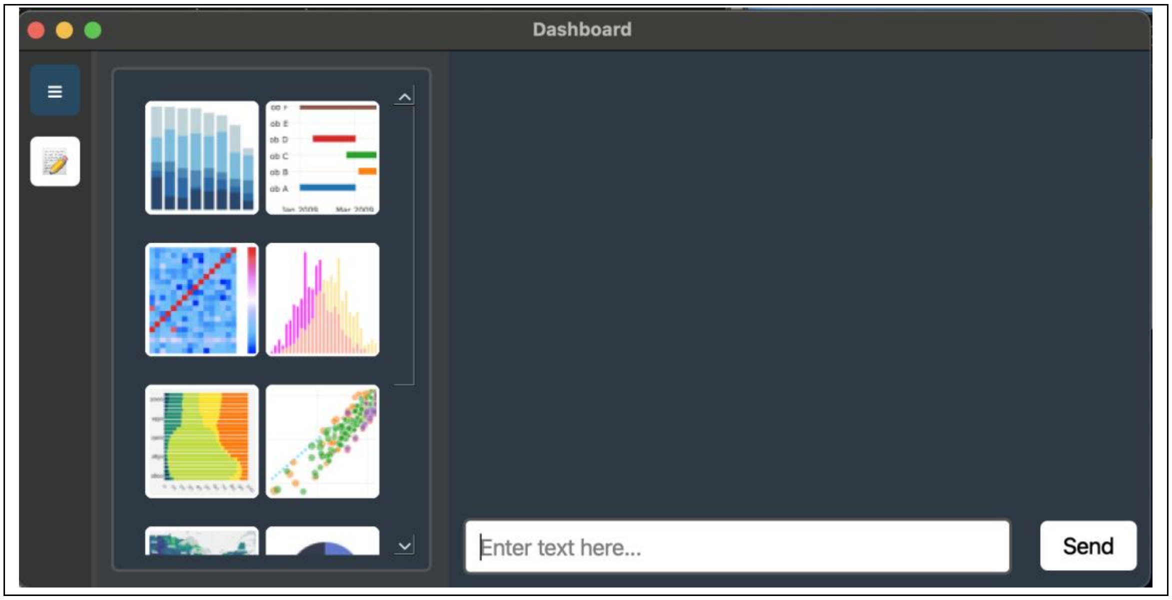


Figure 4.14: Prompt field screen (with sidebar open)

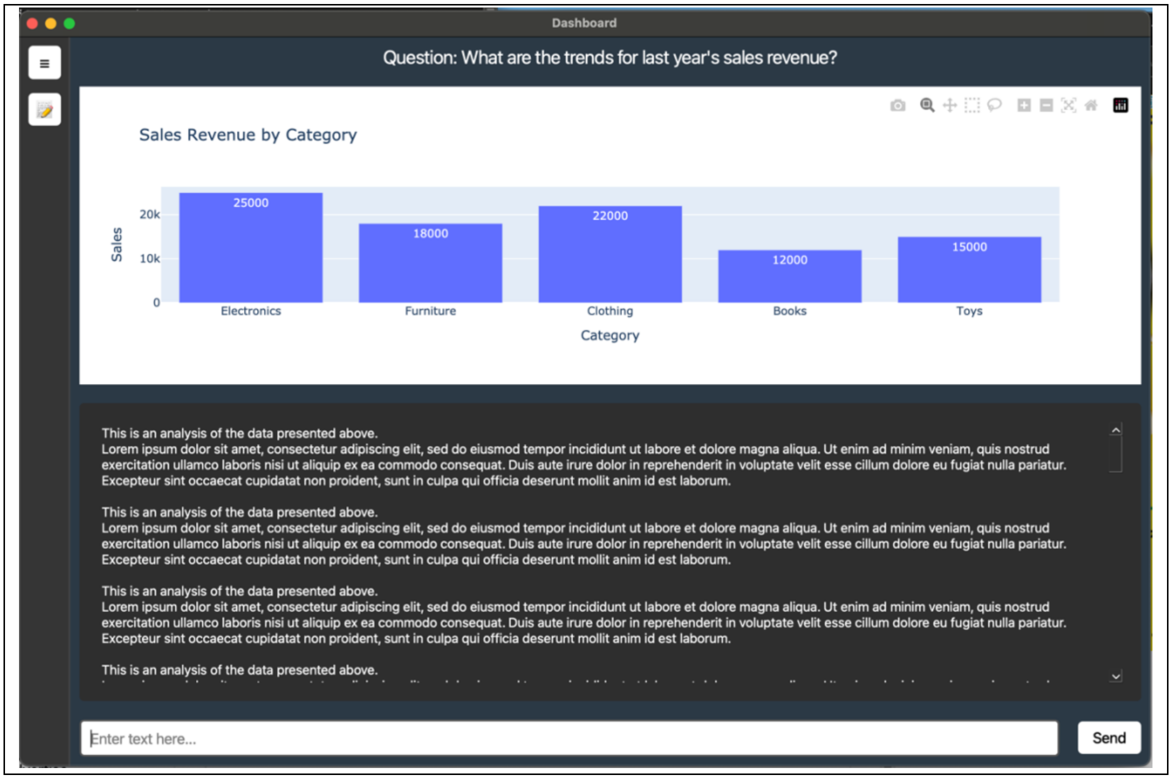


Figure 4.15: Prompt field screen (with visual and analysis)

Chapter 5: Implementation Plan

Based on the project scope, the proposed software development model is the traditional waterfall model of SDLC. The software development life cycle (SDLC) is a process used to design and develop reliable and cost-effective software within a set time. It is also known as the software development process model. SDLC models provide a structured approach to creating high-quality software. As technology changes and market demands grow, organizations must choose the right SDLC model to meet project needs. Developers, project managers, and quality engineers need to understand the benefits and limitations of different models to ensure software quality.

The Waterfall model follows a strict sequence where each phase must be completed before moving to the next. It does not allow returning to a previous phase for changes which makes this suitable for small projects where revisions are minimal. In this model, problems cannot be fixed until the maintenance stage. SDLC models help provide a clear roadmap for software projects by ensuring careful planning and execution at each stage (Pargaonkar, S., 2023). Figure 5.1 displays the waterfall model diagram this project incorporates.

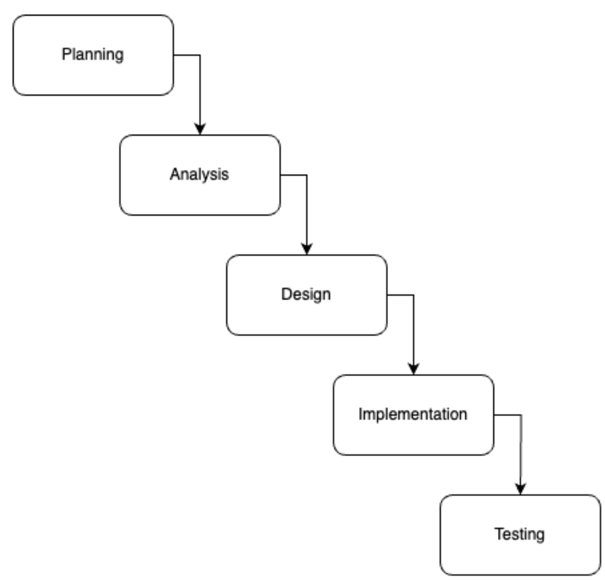


Figure 5.1: Waterfall model diagram (Pargaonkar, S., 2023)

The SDLC begins with planning the baseline and define the scope of the project. The consideration for this project includes the trimester length, classes enrolled and the classwork, the resources, and technical skills.

Afterwards, the analysis phase begins by studying the background of the project. The study done is to introduce theories, concepts, terms, and ideas that may be unfamiliar to the target audience. Concepts such as data visualization, NLP, and LLM are introduced to understand the problems and how these concepts could potentially solve it. Afterward, the project continues by recognizing similar system to the project to gather existing features. The features then are assessed further through questionnaire to further understand which of these features make a proper data visualization tool that follows the current trend. By the end of this phase, the functional and quality requirements of the system are finalized.

The design stage continues from the analysis phase where the high-level design is created to depict the system's functionalities, interaction with external system, the data design, and the function sequence.

The implementation and testing are done to verify and validate the system while at the same time make changes to the system based on the response from the User Acceptance Test (UAT) result. The purpose of these phases is to ensure the system is outcome is as expected or not.

For FYP2, the data visualization tool will be developed. The developed data visualization tool will undergo the testing phase to verify the output, alongside identifying errors generated from different components and functions of the data visualization tool by verifying it with the selected testers. The received feedback will be used to improve the prototype until it reaches the end of the phase where the final product is developed, and the process are properly documented. The presentation for FYP2 of the project will involve the demonstration of the final product. Figure 5.1 visualizes the timeline for FYP2 in the upcoming semester.

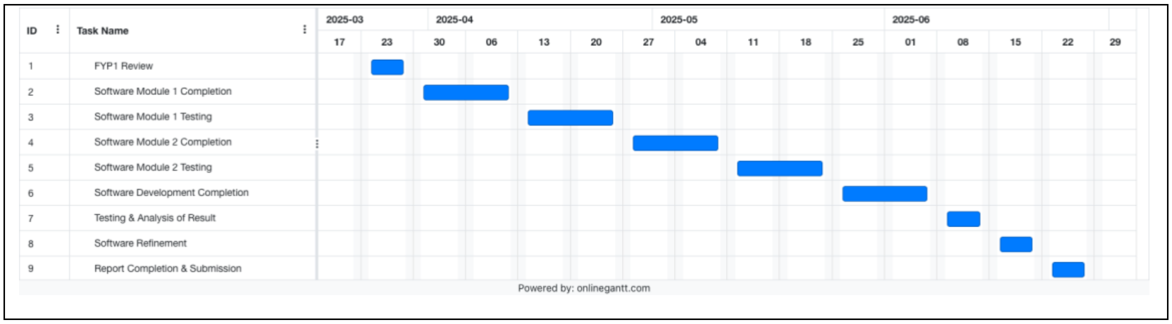


Figure 5.2: FYP2 Gantt chart

Chapter 6: Conclusion

In conclusion, finishing this project's initial phase has expanded my understanding of how technology may be used for good. Given the existence of well-known data visualization tools, it was initially difficult to decide which features and functionalities should be included to make the project stand out. In order to improve usability, it was decided to incorporate Large Language Models (LLMs) into the visualization tool following a full background investigation. The project's next stage will concentrate on creating and perfecting a working prototype. Following the stated timeline and meeting project milestones will be crucial to achieving the best potential result.

References

Al-Fedaghi, S. (2021). UML sequence diagram: an alternative model. *arXiv preprint arXiv:2105.15152*. doi: https://doi.org/10.48550/arXiv.2105.15152

Anisya, A., & Nugroho, F. (2021). Analysis of Back-End Requirements of a Web-Based Application to Identify Damage for Building Structures. *IJISTECH (International Journal of Information System and Technology)*, *5*(4), 431-435.

Argano. (2022, March 21). *What Is Oracle Analytics Cloud? Features, Benefits & More*. Retrieved from https://argano.com/insights/articles/what-is-oracle-analytics-cloud-features-benefits-and-more.html

Biswal, A. (2024, October 24). *What is Power BI?: Architecture, and Features Explained*. Retrieved from https://www.simplilearn.com/tutorials/power-bi-tutorial/what-is-power-bi

Casarotto, C. (2021, May 10). *Google Data Studio: What It Is And How To Use It In* 2024. Retrieved from https://rockcontent.com/blog/how-to-use-google-data-studio/

Choudhary, V. K., Kumar, G., Suman, S., Kumar A. (2024). A Comprehensive Review on Data Visualization Techniques for Real-Time Information. *International Conference on Inventive Computation Technologies (ICICT)* 886-871.

Gobov, D., & Huchenko, I. (2021). Software Requirements Elicitation Techniques Selection Method for the Project Scope Management. *ITPM,* 1-10.

Google LLC. (2025, January 21). *Add charts and controls to your report.* Google Cloud. https://cloud.google.com/looker/docs/studio/add-charts-and-controls-to-your-report?hl=en

Google LLC. (2025, January 30). *Invite others to your reports.* Google Cloud. https://cloud.google.com/looker/docs/studio/invite-others-to-your-reports?hl=en

Google LLC. (2025, January 30). *Welcome to Looker Studio!* Google Cloud. https://cloud.google.com/looker/docs/studio?hl=en

Google LLC. (2025, January 31). *About connectors, data sources, and credentials.* Google Cloud. https://cloud.google.com/looker/docs/studio/about-connectors-data-sources-and-credentials?hl=en

Groq, Inc. (n. d.) *Rate Limits*. GroqCloud Developer Console. https://console.groq.com/docs/rate-limits

Groq, Inc, (n. d.) New AI Inference Speed Benchmark for Llama 3.3 70B, Powered by Groq. Retrieved from https://groq.com/new-ai-inference-speed-benchmark-for-llama-3-3-70b-powered-by-groq

Jinxin, S., Jiabao, Z., Yilei, W., Xingjiao, W., Jiawen, L., & Liang, H. (2023). Cgmi: Configurable general multi-agent interaction framework. *arXiv preprint arXiv:2308.12503*. https://doi.org/10.48550/arXiv.2308.12503

Khurana, D., Koli, A., Khatter, K., & Singh, S. (2023). Natural language processing: state of the art, current trends and challenges. Multimedia tools and applications, 82(3), 3713-3744.

Kircher, R., & Zipp, L. (Eds.). (2022). Questionnaire for elicit Quantitative Data. Research methods in language attitudes (pp. 129-131). Cambridge: Cambridge University Press.

Krüger, G. (2024, August 8). Non-functional Requirements: What They Do, Examples, and Best Practices. Retrieved from https://www.perforce.com/blog/alm/what-are-non-functional-requirements-examples#types-of-non-functional-requirements

Lavanya, A., Sindhuja, S., Gaurav, L., Ali, W. (2023). A Comprehensive Review of Data Visualization Tools: Features, Strengths, and Weaknesses. *International Journal of Computer Engineering in Research Trends, 10*(1), 10-20.

Liang, T., He, Z., Jiao, W., Wang, X., Wang, Y., Wang, R., ... & Tu, Z. (2023). Encouraging divergent thinking in large language models through multi-agent debate. *arXiv preprint arXiv:2305.19118*. doi: https://doi.org/10.48550/arXiv.2305.19118

Mei, K., Li, Z., Xu, S., Ye, R., Ge, Y., & Zhang, Y. (2024). AIOS: LLM agent operating system. *arXiv e-prints*. doi: https://doi.org/10.48550/arXiv.2403.16971

Microsoft Corporation, Inc. (2023, March 09). *Design Power BI reports for accessibility.* Power BI documentation. https://learn.microsoft.com/en-us/power-bi/create-reports/desktop-accessibility-creating-reports

Microsoft Corporation, Inc. (2024, January 24). *Quickstart: Connect to data in Power BI Desktop.* Power BI documentation. https://learn.microsoft.com/en-us/power-bi/connect-data/desktop-quickstart-connect-to-data

Microsoft Corporation, Inc. (2024, March 18). *Share Power BI reports and dashboards with coworkers and others.* Power BI documentation. https://learn.microsoft.com/en-us/power-bi/collaborate-share/service-share-dashboards

Microsoft Corporation, Inc. (2024, March 22). *Create and manage relationships in Power BI Desktop.* Power BI documentation. https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-create-and-manage-relationships

Microsoft Corporation, Inc. (2024, March 22). *What is Power BI?* Microsoft. https://learn.microsoft.com/en-us/power-bi/fundamentals/power-bi-overview

Microsoft Corporation, Inc. (2025, January 06). *Dashboards for business users of the Power BI service.* Power BI documentation. https://learn.microsoft.com/en-us/power-bi/consumer/end-user-dashboards

Mohammadjafari, A., Maida, A. S., & Gottumukkala, R. (2024). From natural language to sql: Review of llm-based text-to-sql systems. *arXiv preprint arXiv:2410.01066*. doi: https://doi.org/10.48550/arXiv.2410.01066

Nurcahya, D., Nurfauziah, H., & Dwiatmodjo, H. (2022). Comparison of Waterfall Models and Prototyping Models of Meeting Management Information Systems. *Jurnal Mantik, 6*(2), 1934-1939.

Oracle Corporation. (n. d.) *Analytics Cloud*. Oracle Cloud Infrastructure Documentation. https://docs.oracle.com/en-us/iaas/analytics-cloud/index.html

Oracle Corporation. (n. d.) *Build Reports and Dashboards*. Oracle Analytics Cloud. https://docs.oracle.com/en/cloud/paas/analytics-cloud/build-reports-and-dashboards.html

Oracle Corporation. (n. d.) *Model Data*. Oracle Analytics Cloud. https://docs.oracle.com/en/cloud/paas/analytics-cloud/model-data-reports.html

Oracle Corporation. (n. d.) *Upload and Connect*. Oracle Analytics Cloud. https://docs.oracle.com/en/cloud/paas/analytics-cloud/upload-data.html

Oracle Corporation. (n. d.) *Visualize Data*. Oracle Analytics Cloud. https://docs.oracle.com/en/cloud/paas/analytics-cloud/visualize-data.html

Pargaonkar, S. (2023). A Comprehensive Research Analysis of Software Development Life Cycle (SDLC) Agile & Waterfall Model Advantages, Disadvantages, and Application Suitability in Software Quality Engineering. *International Journal of Scientific and Research Publications (IJSRP), 13*(08), 345-358.

Qin, L., Chen, Q., Feng, X., Wu, Y., Zhang, Y., Li, Y., ... & Yu, P. S. (2024). Large language models meet nlp: A survey. arXiv preprint arXiv:2405.12819. doi: https://doi.org/10.48550/arXiv.2405.12819

Rashkovits, R., & Lavy, I. (2021). Mapping Common Errors in Entity Relationship Diagram Design of Novice Designers. *International Journal of Database Management Systems, 13*(1), 1-19.

Ruan, J., Chen, Y., Zhang, B., Xu, Z., Bao, T., Mao, H., ... & Zhao, R. (2023). Tptu: Task planning and tool usage of large language model-based ai agents. NeurIPS 2023 Foundation Models for Decision Making Workshop.

Shen, Z. (2024). LLM With Tools: A Survey. arXiv preprint arXiv:2409.18807. doi: https://doi.org/10.48550/arXiv.2409.18807

Skender, F., Manevska, V. (2022). Data Visualization Tools-Preview and Comparison. *Journal of Emerging Computer Technologies, 2*(1), 30-35.

Staff, C. (2024, December 11). *What Is Tableau? Features, Use Cases, and More*. Retrieved from https://www.coursera.org/articles/tableau

Sudiartha, I., Indrayana, I., Suasnawa, I. W., Atmaja, I., Indah, K. A. T., & Sunu, P. W. (2021). User Requirement and Use Case Diagram for Traveler Tracking Application in Tourist Destination. *User Requirement and Use Case Diagram for Traveler Tracking Application in Tourist Destination*, 1, 1376-1380. doi: https://doi.org/10.5220/0010965700003260

Tableau Software, LLC. (n. d.) *Analyze Data.* Tableau Desktop and Web Authoring Help. https://help.tableau.com/current/pro/desktop/en-us/analyze.htm

Tableau Software, LLC. (n. d.) *Automatically Build Views with Ask Data.* Tableau Desktop and Web Authoring Help. https://help.tableau.com/current/pro/desktop/en-us/ask_data.htm

Tableau Software, LLC. (n. d.) *Build Data Views from Scratch.* Tableau Desktop and Web Authoring Help. https://help.tableau.com/current/pro/desktop/en-us/building_overview.htm

Tableau Software, LLC. (n. d.) *Connect to Your Data.* Tableau Desktop and Web Authoring Help. https://help.tableau.com/current/pro/desktop/en-us/basicconnectoverview.htm

Tableau Software, LLC. (n. d.) *Create Dashboards.* Tableau Desktop and Web Authoring Help. https://help.tableau.com/current/pro/desktop/en-us/dashboards.htm

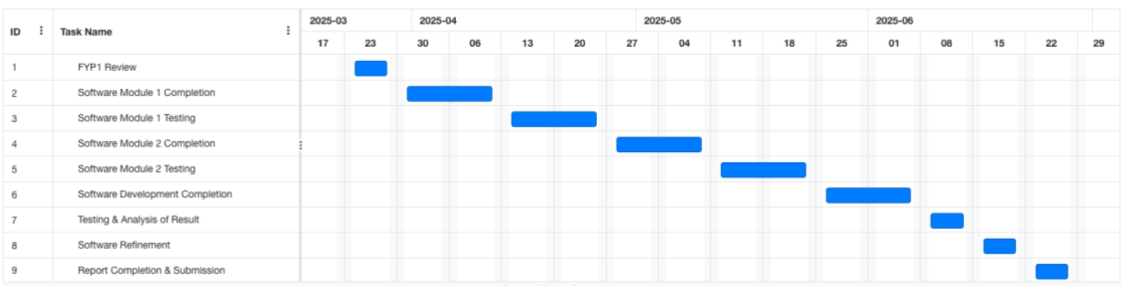
Tableau Software, LLC. (n. d.) *Create Stories.* Tableau Desktop and Web Authoring Help. https://help.tableau.com/current/pro/desktop/en-us/stories.htm

Tableau Software, LLC. (n. d.) *Tableau Product Overview.* Tableau Help. https://help.tableau.com/current/tableau/en-us/tableau_product_overview.htm

Talebirad, Y., & Nadiri, A. (2023). Multi-agent collaboration: Harnessing the power of intelligent llm agents. *arXiv preprint arXiv:2306.03314*. doi: https://doi.org/10.48550/arXiv.2306.03314

Zhao, W. X., Zhou, K., Li, J., Tang, T., Wang, X., Hou, Y., ... & Wen, J. R. (2023). A survey of large language models. *arXiv preprint arXiv:2303.18223*. doi: https://doi.org/10.48550/arXiv.2303.18223

Appendix A: Gantt Chart



Appendix B: FYP I Meeting Logs

CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2024 (Trimester ID:2430)

Meeting Date: 22nd November 2024	**Meeting No.:** 1
Meeting Mode: Online	
Project ID: FYP01-SE-T2430-0033	**Project Type:** Application-based
Project Title : LLM-Powered Dashboard using Multi-Agent Framework	
Student ID : 1211101583	**Student Name:** LUQMAN HAKIM BIN NOORAZMI
Student Programme and Specialisation: Bachelor of Computer Science (Software Engineering)	
Supervisor Name: Dr. Zarina bt Che Embi	**Co-Supervisor Name:** **(if applicable)**
Collaborating Company: **(if applicable)**	**Company Supervisor Name:** **(if applicable)**

1. WORK DONE
[Please write the details of the work done, after the last meeting]

Tasks: Problem Formulation and Project Planning / Background Study or ~~Literature Review~~ / ~~Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept / Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Problem Formulation and Project Planning:
    - Planned the phases of the project for FYP1 and FYP2
- Background Study:
    - Introduced theories, concepts, terms, and ideas that may be unfamiliar to the target audience
    - Introduced concepts that may have been borrowed from other disciplines that may be unfamiliar to the reader that needs explanation
    - Recognized similar system to the project or any existing software that uses the main concepts from the project

2. WORK TO BE DONE
[Please write the details of the work to be done, before the next meeting]

Tasks: ~~Problem Formulation and Project Planning /~~ Background Study ~~or Literature Review /~~ Requirement Analysis ~~or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept/ Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Find research papers related to the system and finalize the background study
- Identify and conduct requirements analysis

3. PROBLEMS ENCOUNTERED AND SOLUTIONS
[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

- **Problem**: Unable to complete part of the abstract before completing the prototype
    - **Solution**: Wait until after the prototype has been developed
- **Problem:** Unable to access some existing dashboard
    - **Solution**: Rely on documentation and Youtube video of it

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

Good

……………………………
Supervisor's Signature

……………………………..
Student's Signature

………………………………
Co-Supervisor's Signature
(if applicable)

…………………………………
Company Supervisor's Signature
(if applicable)

IMPORTANT NOTES TO STUDENTS:

1. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
2. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.
   Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can have fortnightly meetings with the supervisor.
3. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.

   This also provide the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provide the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.
4. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2024 (Trimester ID:2430)

Meeting Date: 5th December 2024	**Meeting No.:** 2
Meeting Mode: Online	
Project ID: FYP01-SE-T2430-0033	**Project Type:** Application-based
Project Title : LLM-Powered Dashboard using Multi-Agent Framework	
Student ID : 1211101583	**Student Name:** LUQMAN HAKIM BIN NOORAZMI
Student Programme and Specialisation: Bachelor of Computer Science (Software Engineering)	
Supervisor Name: Dr. Zarina bt Che Embi	**Co-Supervisor Name:** **(if applicable)**
Collaborating Company: **(if applicable)**	**Company Supervisor Name:** **(if applicable)**

1. WORK DONE
[Please write the details of the work done, after the last meeting]

Tasks: ~~Problem Formulation and Project Planning /~~ Background Study / Requirement Analysis ~~/ Design or Research Methodology /~~
~~Prototype Development or Proof of Concept / Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

Details (in point form):

Background Study

- Dissected the domain of the problem
- Analyzed recent research papers from the domain
- Identified the limitation of the state-of-the-art tools from the domain
- Proposed solutions to the identified limitation

Requirement Analysis

- Based on the tools gathered from the background study, analyzed the various use cases of the tools.

2. WORK TO BE DONE
[Please write the details of the work to be done, before the next meeting]

Tasks: ~~Problem Formulation and Project Planning / Background Study or Literature Review /~~ Requirement Analysis ~~/ Design / Prototype Development or Proof of Concept/ Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Justify the proposed solutions through studies of the domains of the solution
- Organize and analyze the various features from the different tools
- Prepare elicitation techniques to gather requirements

3. PROBLEMS ENCOUNTERED AND SOLUTIONS
[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

- Limited access on some of the tools
    - Rely on any documentation of the tools

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

Good

……………………………
Supervisor's Signature

…………………………...
Student's Signature

………………………………
Co-Supervisor's Signature
(if applicable)

…………………………………
Company Supervisor's Signature
(if applicable)

IMPORTANT NOTES TO STUDENTS:

1. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.

2. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.
   Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can have fortnightly meetings with the supervisor.

3. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.

   This also provide the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provide the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.

4. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2024 (Trimester ID:2430)

Meeting Date: 19th December 2024	**Meeting No.:** 3
Meeting Mode: Online	
Project ID: FYP01-SE-T2430-0033	**Project Type:** Application-based
Project Title : LLM-Powered Dashboard using Multi-Agent Framework	
Student ID : 1211101583	**Student Name:** LUQMAN HAKIM BIN NOORAZMI
Student Programme and Specialisation: Bachelor of Computer Science (Software Engineering)	
Supervisor Name: Dr. Zarina bt Che Embi	**Co-Supervisor Name:** **(if applicable)**
Collaborating Company: **(if applicable)**	**Company Supervisor Name:** **(if applicable)**

1. WORK DONE
[Please write the details of the work done, after the last meeting]

Tasks: ~~Problem Formulation and Project Planning / Background Study /~~ Requirement Analysis ~~/ Design or Research Methodology /~~
~~Prototype Development or Proof of Concept / Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

Details (in point form):

Requirement Analysis

- Justified the proposed solutions through studies of the domains of the solution
- Organized and analyzed various features from different tools
- Prepared elicitation techniques to gather requirements

2. WORK TO BE DONE
[Please write the details of the work to be done, before the next meeting]

Tasks: ~~Problem Formulation and Project Planning / Background Study or Literature Review /~~ Requirement Analysis / Design ~~/ Prototype Development or Proof of Concept/ Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

Details (in point form):

Requirement Analysis

- Execute the elicitation process
- Finalize the functional and non-functional requirements

Design

- Define the use cases and architecture of the project

3. PROBLEMS ENCOUNTERED AND SOLUTIONS
[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

- Some identified existing systems from the research changed its name.
  - Verified for any feature changes between the one from the research (old name) with the latest system's feature (new name)
- Lack of elaboration from the authors regarding the existing system's limitation
  - Went through multiple references the limitations were sourced from

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)
Good

…………………………… Supervisor's Signature

…………………………... Student's Signature

……………………………. Co-Supervisor's Signature **(if applicable)**

…………………………………. Company Supervisor's Signature **(if applicable)**

IMPORTANT NOTES TO STUDENTS:

1. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.

2. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.
   Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can have fortnightly meetings with the supervisor.

3. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.

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4. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2024 (Trimester ID:2430)

Meeting Date: 2nd January 2025	**Meeting No.:** 4
Meeting Mode: Online	
Project ID: FYP01-SE-T2430-0033	**Project Type:** Application-based
Project Title : LLM-Powered Dashboard using Multi-Agent Framework	
Student ID : 1211101583	**Student Name:** LUQMAN HAKIM BIN NOORAZMI
Student Programme and Specialisation: Bachelor of Computer Science (Software Engineering)	
Supervisor Name: Dr. Zarina bt Che Embi	**Co-Supervisor Name:** **(if applicable)**
Collaborating Company: **(if applicable)**	**Company Supervisor Name:** **(if applicable)**

1. WORK DONE
[Please write the details of the work done, after the last meeting]

Tasks: ~~Problem Formulation and Project Planning / Background Study /~~ Requirement Analysis / ~~Design / Prototype Development or Proof of Concept / Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

Details (in point form):
Requirement Analysis

- Executed the elicitation process
- Finalized the functional requirements
- Defined some of the quality requirements
- Defined the requirements in terms of context diagram and level 0 DFD
- Defined the use cases and the high-level architecture of the system

2. WORK TO BE DONE
[Please write the details of the work to be done, before the next meeting]

Tasks: ~~Problem Formulation and Project Planning / Background Study or Literature Review /~~ Requirement Analysis / Design ~~/ Prototype Development or Proof of Concept/ Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

Details (in point form):
Requirement Analysis

- Continue defining the quality requirements
- Describe and organize all the identified use cases (actors, scenarios, pre/postconditions)
- Start writing report for chapter 3

Design
- Create the sequence diagram using Visual Paradigm software
- Design the system interface

3. PROBLEMS ENCOUNTERED AND SOLUTIONS
[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

- Was not sure whether the use of LLM counts as external system or not
  - Since it is integrated into the system, it is decided to be counted as the system, even though it's still dependent on external resource to get the model
- Hard time deciding the metrics for the non-functional requirements
  - Do thorough research on existing model's response time, converting text to SQL, executing query and getting the data (might need more time for large database,

should consider limiting the amount of data from the db), and time for the model to process the data.
4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor) Good

……………………………
Supervisor's Signature

…………………………...
Student's Signature

……………………………..
Co-Supervisor's Signature
(if applicable)

…………………………………
Company Supervisor's Signature
(if applicable)

IMPORTANT NOTES TO STUDENTS:

1. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
2. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.
   Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can have fortnightly meetings with the supervisor.
3. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.

   This also provide the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provide the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.
4. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

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CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2024 (Trimester ID:2430)

Meeting Date: 16th January 2025	**Meeting No.:** 5
Meeting Mode: Online	
Project ID: FYP01-SE-T2430-0033	**Project Type:** Application-based
Project Title : LLM-Powered Dashboard using Multi-Agent Framework	
Student ID : 1211101583	**Student Name:** LUQMAN HAKIM BIN NOORAZMI
Student Programme and Specialisation: Bachelor of Computer Science (Software Engineering)	
Supervisor Name: Dr. Zarina bt Che Embi	**Co-Supervisor Name:** **(if applicable)**
Collaborating Company: **(if applicable)**	**Company Supervisor Name:** **(if applicable)**

1. WORK DONE
[Please write the details of the work done, after the last meeting]

Tasks: ~~Problem Formulation and Project Planning / Background Study /~~ Requirement Analysis / Design ~~/ Prototype Development or Proof of Concept / Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

Details (in point form):

Requirement Analysis
- Defined the quality requirements
- Described and organized all the identified use cases (actors, scenarios, pre/postconditions)
- Started to design the report for chapter 3
- Modified the context diagram

Design
- Designed the sequence diagram for every use cases
- Started to design the system backend as part of designing the interface

2. WORK TO BE DONE
[Please write the details of the work to be done, before the next meeting]

Tasks: ~~Problem Formulation and Project Planning / Background Study or Literature Review /~~ Requirement Analysis / Design / Prototype Development or Proof of Concept / Draft Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

Requirement Analysis
- Rework the context diagram

Design
- Start to design the system data
- Continue to design the system interface

Prototype Development
- Start to develop the system prototype

Draft Report Completion
- Continue to write report for chapter 3
- Continue to write report for chapter 4

3. PROBLEMS ENCOUNTERED AND SOLUTIONS *[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]* • The are errors found on the context diagram   o Received suggestions and modify the diagram accordingly
4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor) Good

…………………………… ……………………………..
Supervisor's Signature Student's Signature

……………………………… …………………………………
Co-Supervisor's Signature **(if applicable)** Company Supervisor's Signature **(if applicable)**

IMPORTANT NOTES TO STUDENTS:

1. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.

2. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.
   Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can have fortnightly meetings with the supervisor.

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4. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2024 (Trimester ID:2430)

Meeting Date: 4th February 2025	**Meeting No.:** 6
Meeting Mode: Online	
Project ID: FYP01-SE-T2430-0033	**Project Type:** Application-based
Project Title : LLM-Powered Dashboard using Multi-Agent Framework	
Student ID : 1211101583	**Student Name:** LUQMAN HAKIM BIN NOORAZMI
Student Programme and Specialisation: Bachelor of Computer Science (Software Engineering)	
Supervisor Name: Dr. Zarina bt Che Embi	**Co-Supervisor Name:** **(if applicable)**
Collaborating Company: **(if applicable)**	**Company Supervisor Name:** **(if applicable)**

1. WORK DONE
[Please write the details of the work done, after the last meeting]

Tasks: ~~Problem Formulation and Project Planning / Background Study /~~ Requirement Analysis / Design / Prototype Development or Proof of Concept / Draft Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

Requirement Analysis
- Reworked the context diagram and use case diagram

Design
- Designed the system data
- Designed the system interface

Prototype development
- Developed the system prototype based on its design

Draft Report Completion
- Finished writing report for chapter 1-6

2. WORK TO BE DONE
[Please write the details of the work to be done, before the next meeting]

Tasks: ~~Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis / Design /~~ Prototype Development or Proof of Concept / Draft Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

Prototype Development

- Finish developing the prototype based on the system data and system interface

Draft Report Completion

- Finalize the content of the report
- Finish writing report with proper formatting

3. PROBLEMS ENCOUNTERED AND SOLUTIONS
[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions] ○ Minor errors on the report content and report formatting error ▪ Make adjustments and changes to the report based on the error found
4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor) Need to expedite the deliverables.

……………………
Supervisor's Signature

……………………………..
Student's Signature

………………………………
Co-Supervisor's Signature
(if applicable)

…………………………………
Company Supervisor's Signature
(if applicable)

IMPORTANT NOTES TO STUDENTS:

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4. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

Appendix C: Turnitin Similarity Index Page

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File size:	5.52M
Page count:	116
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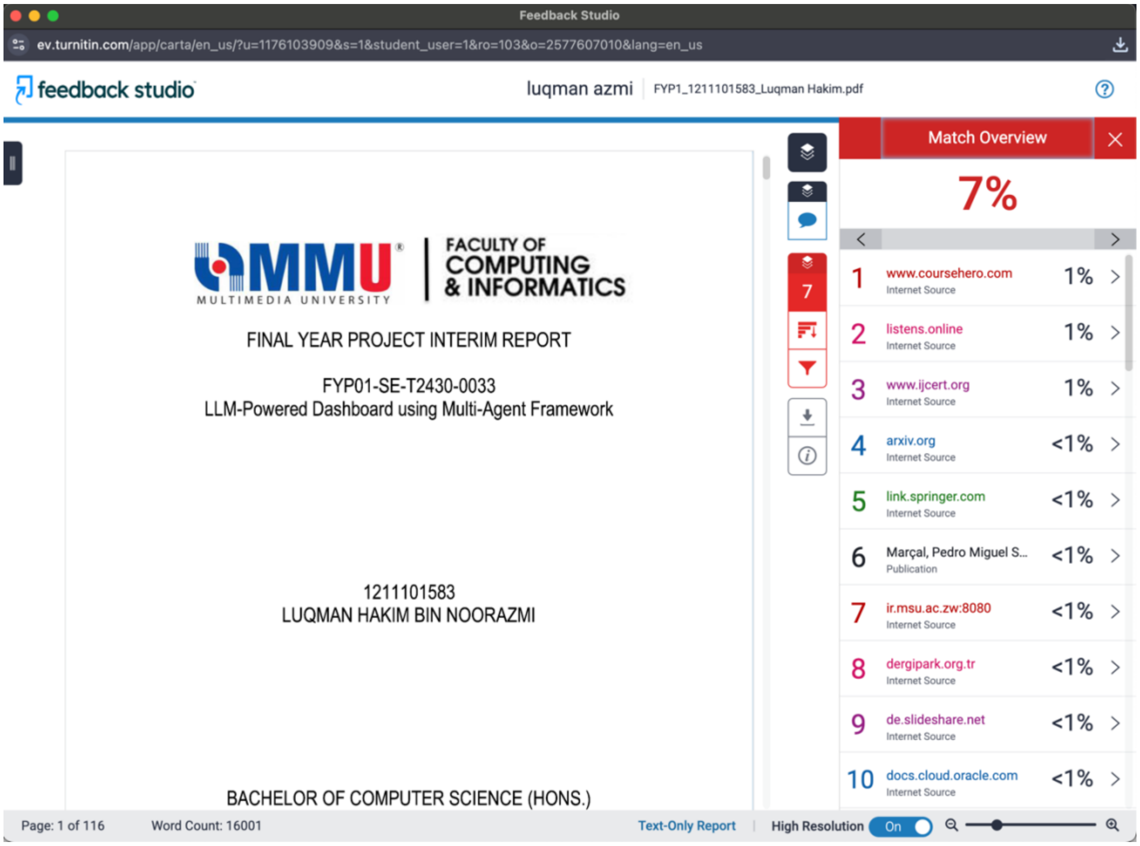
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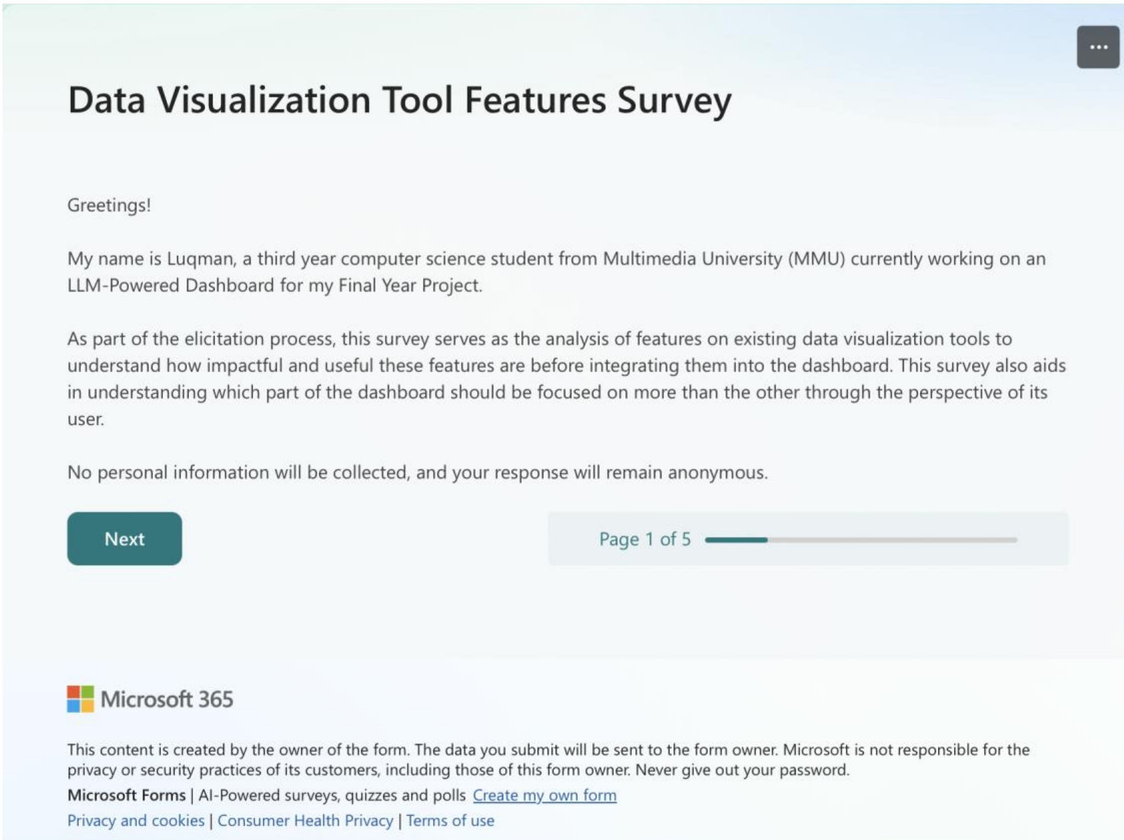
Appendix D: Questionnaire Design



Data Visualization Tool Features Survey

Greetings!

My name is Luqman, a third year computer science student from Multimedia University (MMU) currently working on an LLM-Powered Dashboard for my Final Year Project.

As part of the elicitation process, this survey serves as the analysis of features on existing data visualization tools to understand how impactful and useful these features are before integrating them into the dashboard. This survey also aids in understanding which part of the dashboard should be focused on more than the other through the perspective of its user.

No personal information will be collected, and your response will remain anonymous.

Next

Page 1 of 5

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Data Visualization Tool Features Survey

Background Check

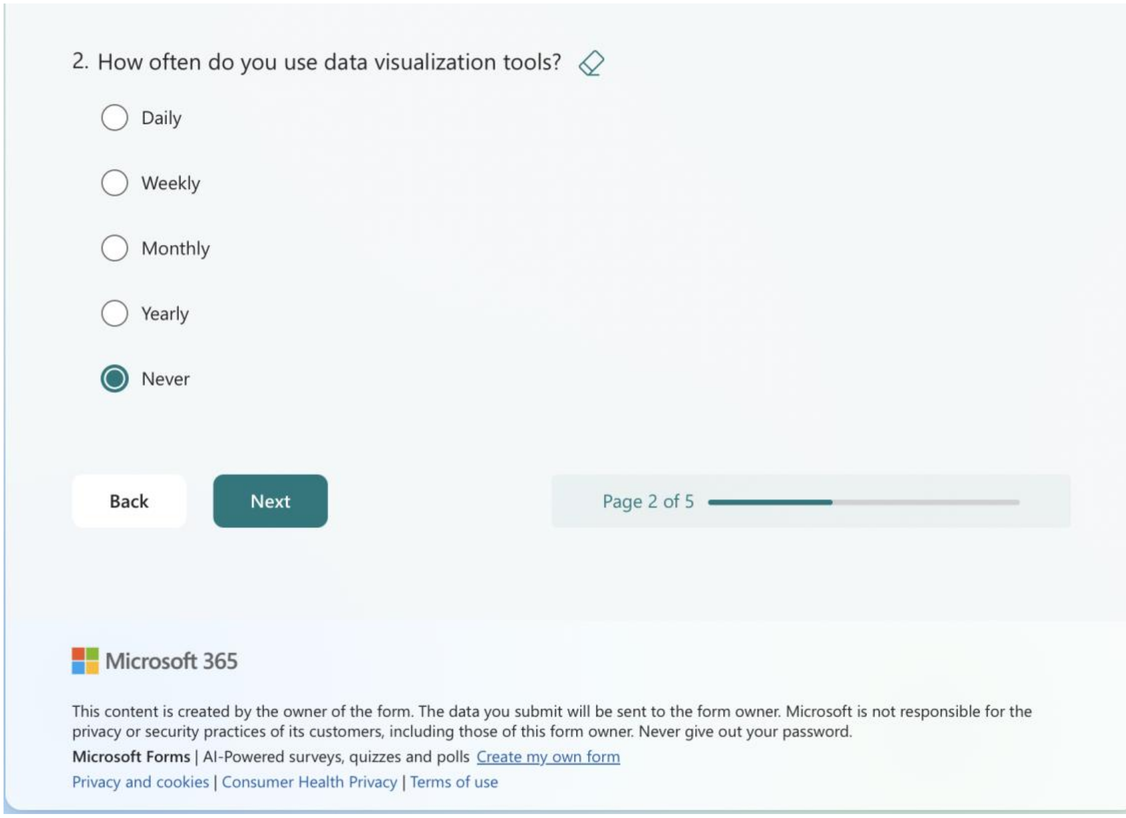
This section is to help identify the respondents background to ensure the survey results reflect the right group of target user. The responses from this section will be used to help confirm that the survey is reaching the intended audience and can be adjusted to better suit different demographic groups if needed.

1. What is your professional status?

- ○ Data Analyst
- ○ Business Intelligence Professional
- ○ Software Developer
- ○ Student
- ○ Other

2. How often do you use data visualization tools?

- ○ Daily
- ○ Weekly
- ○ Monthly


2. How often do you use data visualization tools?

- Daily
- Weekly
- Monthly
- Yearly
- Never

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Page 2 of 5

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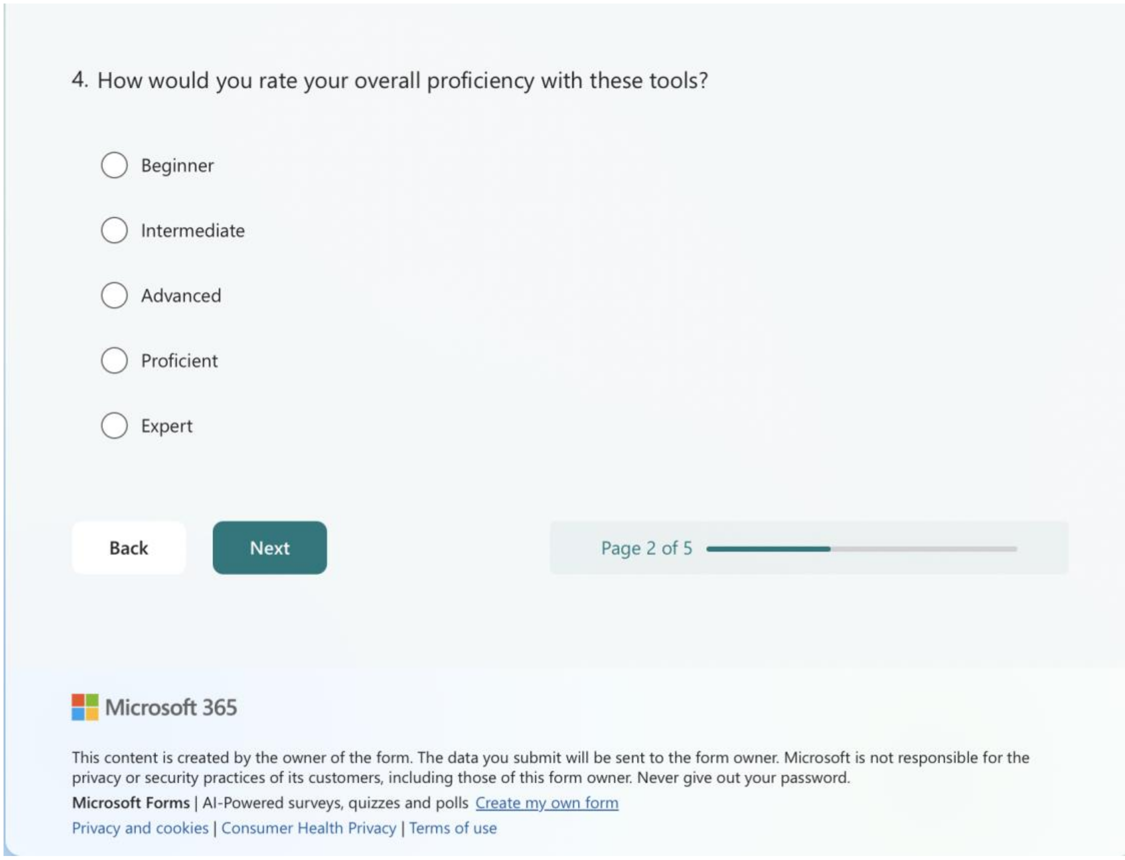
2. How often do you use data visualization tools?

- ◉ Daily
- ○ Weekly
- ○ Monthly
- ○ Yearly
- ○ Never

3. Which of the following data visualization tools have you used? (Select all that apply)

- ☐ Power BI
- ☐ Tableau
- ☐ Looker Studio
- ☐ Oracle Analytics Cloud
- ☐ Other



4. How would you rate your overall proficiency with these tools?

- Beginner
- Intermediate
- Advanced
- Proficient
- Expert

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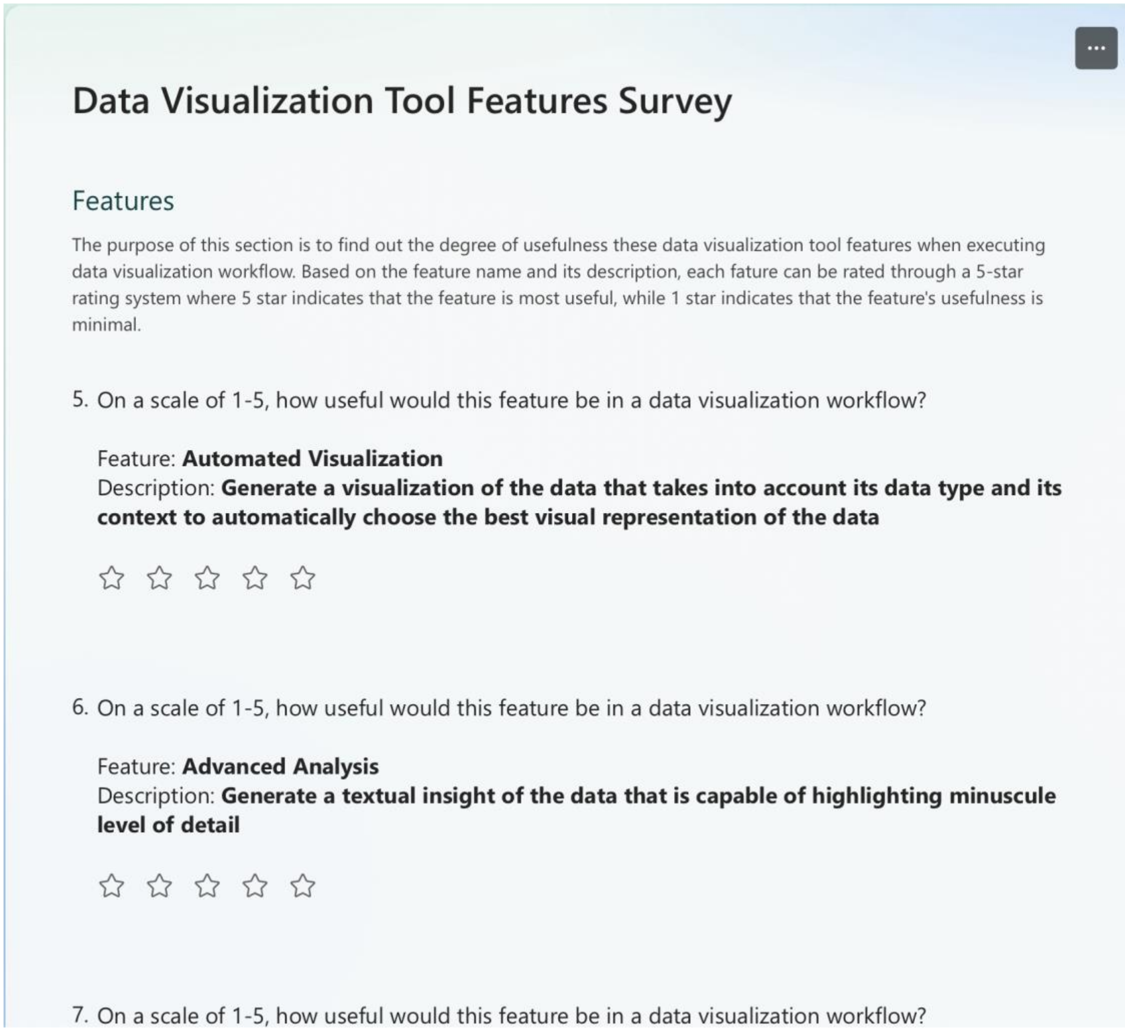

Data Visualization Tool Features Survey

Features

The purpose of this section is to find out the degree of usefulness these data visualization tool features when executing data visualization workflow. Based on the feature name and its description, each fature can be rated through a 5-star rating system where 5 star indicates that the feature is most useful, while 1 star indicates that the feature's usefulness is minimal.

5. On a scale of 1-5, how useful would this feature be in a data visualization workflow?

   Feature: **Automated Visualization**
   Description: **Generate a visualization of the data that takes into account its data type and its context to automatically choose the best visual representation of the data**

6. On a scale of 1-5, how useful would this feature be in a data visualization workflow?

   Feature: **Advanced Analysis**
   Description: **Generate a textual insight of the data that is capable of highlighting minuscule level of detail**

7. On a scale of 1-5, how useful would this feature be in a data visualization workflow?

5. On a scale of 1-5, how useful would this feature be in a data visualization workflow?

   Feature: **Data Storytelling**
   Description: **Provide a narrative insight into the data to relate the trends with the outcome**

6. On a scale of 1-5, how useful would this feature be in a data visualization workflow?

   Feature: **Data Modelling**
   Description: **A set of tools to assist in translating data relationship and data flow into diagram**

7. On a scale of 1-5, how useful would this feature be in a data visualization workflow?

   Feature: **Collaboration**
   Description: **Collaborate with other user in generating data visualization**

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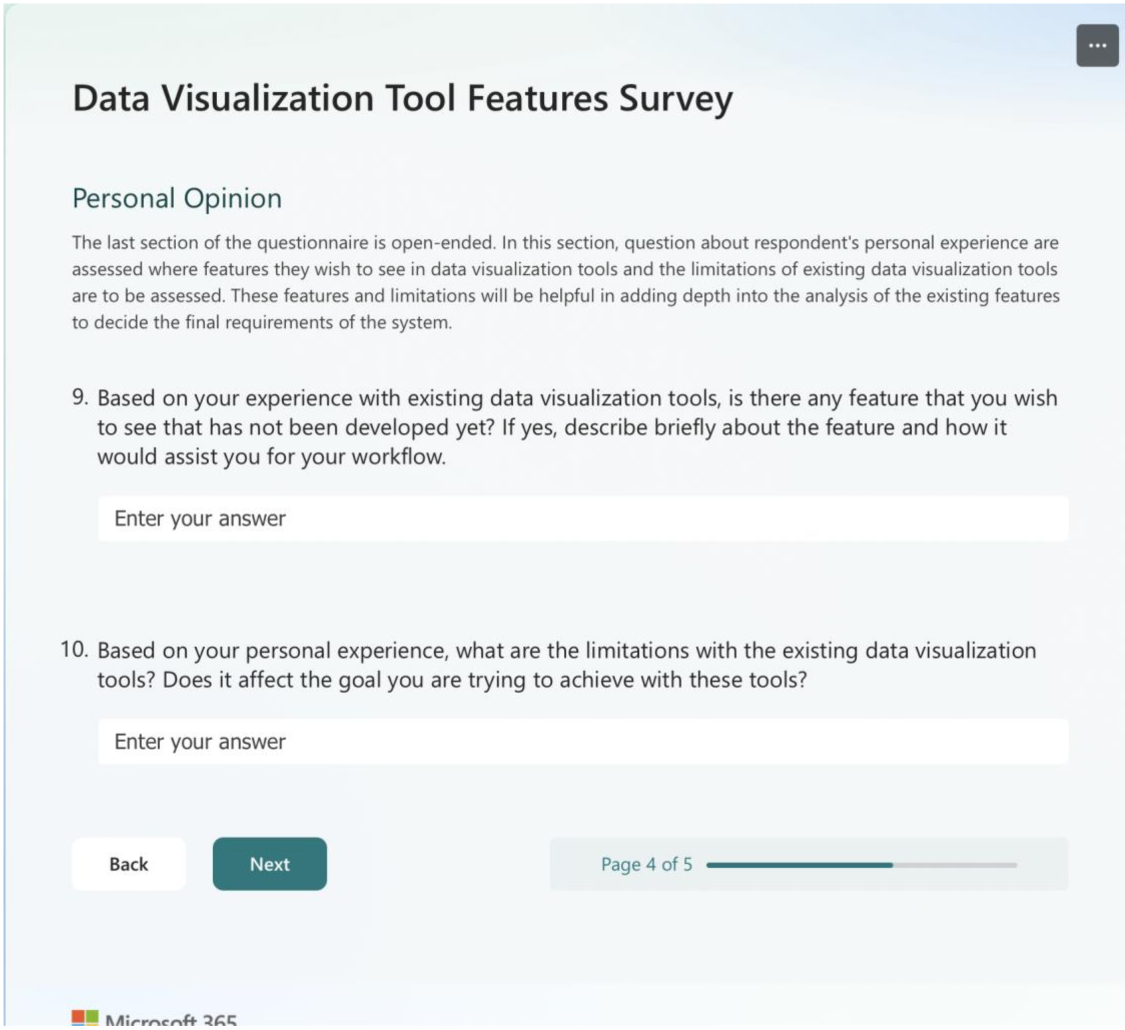

Data Visualization Tool Features Survey

Personal Opinion

The last section of the questionnaire is open-ended. In this section, question about respondent's personal experience are assessed where features they wish to see in data visualization tools and the limitations of existing data visualization tools are to be assessed. These features and limitations will be helpful in adding depth into the analysis of the existing features to decide the final requirements of the system.

9. Based on your experience with existing data visualization tools, is there any feature that you wish to see that has not been developed yet? If yes, describe briefly about the feature and how it would assist you for your workflow.

Enter your answer

10. Based on your personal experience, what are the limitations with the existing data visualization tools? Does it affect the goal you are trying to achieve with these tools?

Enter your answer

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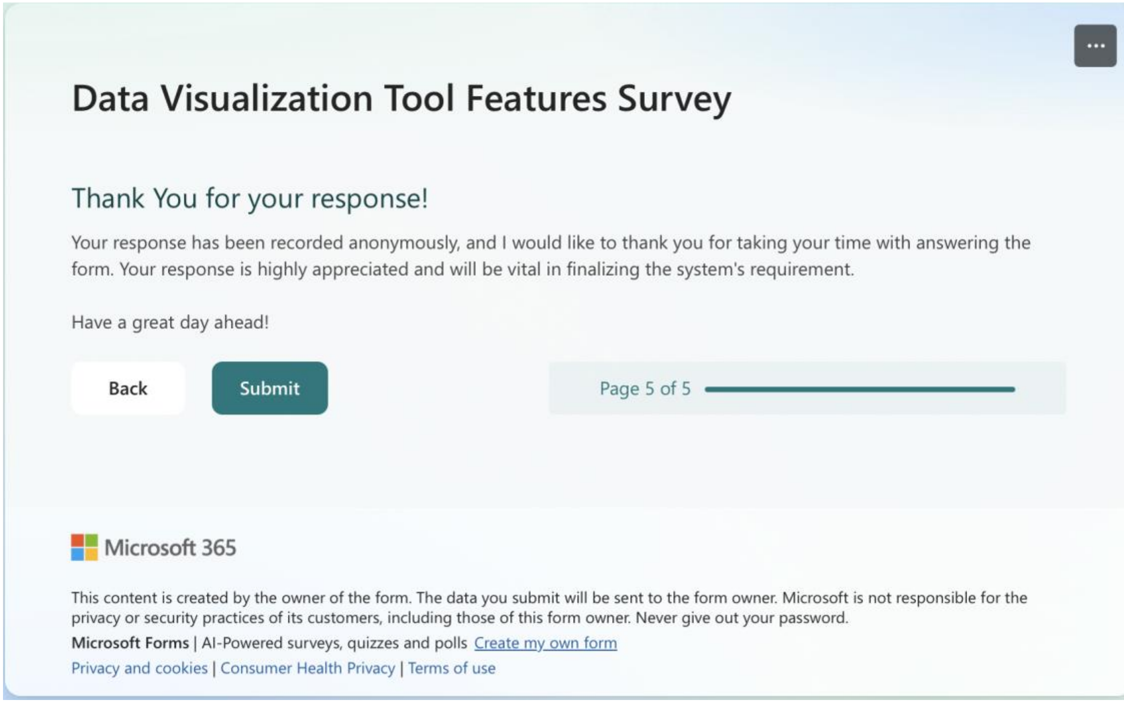

Data Visualization Tool Features Survey

Thank You for your response!

Your response has been recorded anonymously, and I would like to thank you for taking your time with answering the form. Your response is highly appreciated and will be vital in finalizing the system's requirement.

Have a great day ahead!

Back Submit Page 5 of 5

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